

AI-Powered Oral Exam Platform for Scalable and Valid Academic Assessment

Applied Project Final Report

By

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Declaration

I, Danxia Hong, declare that this project report submitted by me to School of Professional Studies, New York University in partial fulfillment of the requirement for the award of the degree of Master of Science in Management and Analytics is a record of project work carried out by me under the guidance of Dr. Andres Fortino, NYU Clinical Assistant Professor of Management and Analytics. I grant powers of discretion to the Division of Programs in Business, School of Professional Studies, and New York University to allow this report to be copied in part or in full without further reference to me. The permission covers only copies made for study purposes or for inclusion in Division of Programs in Business, School of Professional Studies, and New York University research publications, subject to normal conditions of acknowledgment. I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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Abstract

The rapid adoption of generative artificial intelligence in education has introduced both opportunities and challenges in academic assessment, particularly in ensuring the validity and integrity of student evaluations. This project addresses these challenges by proposing an AI-powered oral examination platform designed to deliver scalable, interactive, and reliable student assessments. The proposed solution leverages large language models, such as those developed by OpenAI, combined with voice-based interaction technologies to simulate real-time oral exams. The platform enables instructors to configure customized assessments aligned with course objectives, while students engage in dynamic spoken dialogue with an AI examiner. A multi-agent grading framework, referred to as a “council-of-LLMs,” is implemented to ensure consistency, transparency, and robustness in evaluation. The proof of concept aims to demonstrate the feasibility of automating oral assessments while maintaining assessment quality comparable to traditional methods. The system was evaluated through simulated exam scenarios, focusing on grading consistency, response relevance, and overall user experience. Experimental results indicate that the platform can effectively assess higher-order thinking skills while significantly reducing instructor workload and improving scalability. The findings suggest that AI-powered oral exams can serve as a viable alternative or complement to traditional assessments, particularly in large-scale or online learning environments. However, limitations related to model bias, response variability, and dependence on prompt design remain. Future work will focus on improving evaluation accuracy, integrating adaptive questioning strategies, and conducting real-world pilot studies. This project contributes to the field of AI in education by presenting a scalable framework for automated oral assessment and demonstrating the practical application of large language models in enhancing academic evaluation systems.

Keywords: AI-powered assessment; oral examination; large language models; voice interaction; automated grading; multi-agent systems; academic integrity; educational technology

Abbreviations and Definitions

Abbreviation	Full Term	Definition
AI	Artificial Intelligence	The simulation of human intelligence processes by machines, especially computer systems.
LLM	Large Language Model	A type of AI model trained on large datasets to understand and generate human-like language.
NLP	Natural Language Processing	A field of AI focused on the interaction between computers and human language.
STT	Speech-to-Text	Technology that converts spoken language into written text.
TTS	Text-to-Speech	Technology that converts written text into spoken audio output.
MVP	Minimum Viable Product	A version of a product with sufficient features to validate a concept.
API	Application Programming Interface	A set of rules that allows different software systems to communicate with each other.
LLM Council	Council of Large Language Models	A multi-agent framework where multiple LLMs collaborate to evaluate responses and improve grading reliability.

Introduction

The rapid advancement of large language models (LLMs) has created new opportunities for automating and improving complex business and educational processes. In particular, LLM-powered systems have demonstrated strong capabilities in natural language understanding, reasoning, and interactive dialogue, enabling more dynamic and personalized user experiences. As organizations seek to improve efficiency and scalability, these technologies are increasingly being explored for applications beyond traditional text-based automation.

One critical area where these advancements can be applied is academic assessment. Traditional written examinations often fail to fully capture a student's depth of understanding and are vulnerable to academic integrity issues, particularly in online or remote learning environments. Instructors face increasing challenges in evaluating higher-order thinking skills while maintaining fairness, consistency, and scalability across large student populations.

Problem

The specific problem addressed in this project is the limitation of existing assessment methods in accurately evaluating authentic student understanding at scale. Written exams and automated grading systems primarily assess memorization and structured responses, while oral exams, though more effective at evaluating conceptual understanding, are resource-intensive and difficult to scale. Additionally, the rise of generative AI tools has further complicated academic integrity, making it harder to distinguish between original student work and AI-assisted responses.

Approach

To address these challenges, this project proposes the development of an AI-powered oral examination platform that leverages voice-based interaction and large language models to simulate real-time oral assessments. The system enables instructors to configure customized oral exams aligned with course objectives while allowing students to respond dynamically through spoken interaction. A multi-agent grading framework is incorporated to evaluate responses in a structured, transparent, and consistent manner.

Core Technology

The core technology underlying this project is large language models, such as those developed by OpenAI, which are capable of generating human-like dialogue and performing complex reasoning tasks. These models are integrated with speech-to-text and text-to-speech systems to enable seamless voice interaction between the student and the AI examiner.

In addition, the platform utilizes a multi-agent architecture, where multiple LLM instances collaboratively evaluate student responses. This "council-of-LLMs" approach improves grading reliability by reducing individual model bias and increasing consistency across

evaluations. Prompt engineering techniques are used to guide both questioning strategies and grading criteria, ensuring alignment with academic standards.

Benefits

The proposed solution offers several key benefits. First, it enhances assessment quality by enabling evaluation of deeper conceptual understanding through interactive questioning. Second, it improves scalability by automating the oral examination process, allowing instructors to assess a larger number of students without proportional increases in workload. Third, it strengthens academic integrity by requiring real-time verbal responses, which are more difficult to outsource or generate using external tools. Finally, it provides a more engaging and personalized assessment experience for students.

Research Question

The central research question explored in this project is: Can an AI-powered oral examination platform effectively replicate the benefits of traditional oral exams while achieving scalability, consistency, and reliability in academic assessment?

This paper is organized as follows. The next section defines the project objectives and evaluation metrics. This is followed by an analysis of alternative solutions and a review of relevant literature. The methodology section describes the system design, experimental setup, and evaluation approach. The results section presents findings from the proof of concept, followed by a discussion of issues encountered and lessons learned. The report concludes with an assessment of the project's outcomes and directions for future work.

Contribution

This project contributes to the growing body of research on AI applications in education by demonstrating a scalable and practical implementation of LLM-powered oral assessment. It introduces a novel integration of voice interaction and multi-agent grading within an academic evaluation context and provides empirical insights into the feasibility of replacing or augmenting traditional assessment methods with AI-driven solutions.

Sponsor

The project was conducted under the guidance of Dr. Andres Fortino, Clinical Assistant Professor of Management and Analytics at New York University's School of Professional Studies. The sponsor operates within the higher education sector, with a focus on integrating emerging technologies into business education and assessment practices.

Importance of Project

The successful implementation of this project has significant implications for the sponsor and the broader academic community. As class sizes increase and online learning becomes more prevalent, there is a growing need for scalable and reliable assessment solutions. This project addresses these needs by providing a practical framework for automating oral exams while

maintaining assessment quality. For the sponsor, the platform offers a pathway to enhance teaching effectiveness, improve student evaluation, and position the institution at the forefront of AI-driven educational innovation.

Project Objectives and Metrics

Goal of the project

The goal of this project is to design and develop a scalable, AI-powered oral examination platform that enables faculty to assess genuine student understanding through voice-based interaction. The platform will allow instructors to configure oral exams without custom coding, tailoring questions to course objectives and individual student context. By leveraging voice AI and multi-agent workflows, the system will conduct personalized oral assessments at scale. A council-of-LLMs grading framework will be implemented to ensure consistent, transparent, and evidence-based evaluation. The project aims to deliver a deployable prototype that can be used across multiple courses at NYU SPS within a single academic semester.

Project Deliverables and Metrics

Project Objective 1 – Define the scope, requirements, and system architecture for the AI-powered oral exam platform, including exam configuration, student context inputs, grading workflows, and compliance considerations.

Deliverable: Platform requirements and system design document

Metric: Sponsor approval of requirements and architecture by mid-semester

Project Objective 2 – Develop a working prototype of the AI-powered oral exam platform capable of conducting end-to-end voice-based oral exams with configurable topics and automated grading.

Deliverable: Working MVP of the oral exam platform

Metric: Successful live demonstration of a complete oral exam workflow to Professor Fortino

Project Objective 3 – Pilot the platform using representative course scenarios to validate grading consistency, assessment validity, and student experience compared to traditional written assessments.

Deliverable: Pilot evaluation and validation report

Metric: Sponsor acceptance of validation results and grading reliability

Project Objective 4 – Finalize the platform for deployment by delivering documentation, configuration guidelines, analytics dashboards, and instructor support materials.

Deliverable: Deployment-ready platform package, including a documented GitHub repository and final project report

Metric: Sponsor sign-off and project completion by end of semester

Project Evaluation

The project objectives were evaluated based on the performance of the AI-powered oral examination prototype in a simulated testing environment. Overall, the results indicate that the system successfully met its primary goals, with some limitations typical of an early-stage prototype.

First, the objective of enabling automated oral assessments was achieved. The system was able to simulate an interactive oral examination process, where questions were presented to the student and responses were evaluated using a large language model. The interaction flow demonstrated that the platform can replicate key elements of traditional oral exams in a digital format.

Second, the objective of providing consistent grading was largely met. Through repeated evaluations of selected responses, the system showed relatively low variation in scores, indicating a reasonable level of grading consistency. While minor differences were observed across repeated evaluations, these variations remained within an acceptable range and did not significantly impact overall assessment outcomes.

Third, the system demonstrated the ability to differentiate between varying levels of student performance. High-quality responses consistently received higher scores, while low-quality responses received lower scores. Students with mixed-quality responses obtained correspondingly varied scores across questions. This suggests that the grading mechanism is sensitive to differences in response quality and can approximate human-like evaluation patterns.

In terms of efficiency, the system significantly reduced the time required for grading compared to traditional manual evaluation. While a human grader may take several minutes to assess a single response, the AI system was able to generate scores within seconds, demonstrating strong potential for scalability in large classroom settings.

However, not all objectives were fully achieved. The current prototype requires exam questions to be defined within the application code, which limits flexibility for non-technical users. Additionally, the grading process remains dependent on prompt design, which may introduce variability in certain edge cases. These limitations highlight areas for future improvement.

Overall, the project met its core objectives by demonstrating the feasibility of an AI-powered oral examination system. The results support the potential of such systems to enhance assessment quality, improve scalability, and address challenges in academic integrity, while also identifying opportunities for further refinement.

Alternate Solutions Evaluated

Several alternative approaches to student assessment were considered before selecting the AI-powered oral examination platform as the proposed solution. To ensure a structured and objective decision-making process, a set of evaluation criteria was established and applied consistently across all alternatives.

Solution Evaluation Criteria

The following criteria were used to evaluate and compare the different assessment solutions:

- **Assessment Quality:** The ability to evaluate higher-order thinking, conceptual understanding, and reasoning skills rather than simple memorization.
- **Scalability:** The capacity to support large numbers of students without significantly increasing instructor workload.
- **Academic Integrity:** The ability to reduce opportunities for cheating or unauthorized use of external tools, particularly in remote or online environments.
- **Flexibility:** The extent to which the system can support different types of questions and adapt to various subjects or course requirements.
- **Efficiency:** The time and effort required for grading and administering assessments.

These criteria were selected to reflect both the academic requirements of effective assessment and the operational needs of modern educational environments.

Evaluation of Alternative Solutions

1. Traditional Written Examinations

Traditional written exams perform well in terms of scalability and ease of administration. However, they are limited in assessing deep conceptual understanding and are increasingly vulnerable to academic integrity issues due to the widespread availability of generative AI tools. While efficient, they do not effectively measure real-time reasoning or student authenticity.

2. Automated Grading Systems

Automated grading systems, such as multiple-choice assessments or rule-based evaluation tools, provide high efficiency and scalability. They allow instructors to evaluate large numbers of students quickly and consistently. However, these systems lack flexibility and are limited in assessing open-ended responses and critical thinking. As a result, they perform poorly in terms of assessment quality.

3. AI-Assisted Written Grading

AI-assisted grading of written responses offers improved flexibility compared to traditional automated systems. Large language models can evaluate essays and short answers, enabling more nuanced assessment. However, this approach still relies on written input, which remains

susceptible to AI-assisted cheating. Therefore, while it improves scalability and efficiency, it does not fully address academic integrity concerns.

Selection Rationale

Based on the evaluation criteria, the AI-powered oral examination platform was selected as the preferred solution because it provides a balanced improvement across all key dimensions.

First, in terms of assessment quality, oral examinations allow for dynamic interaction and follow-up questioning, enabling deeper evaluation of student understanding. The integration of large language models enhances this capability by generating context-aware questions and evaluating responses in real time.

Second, the proposed solution significantly improves scalability compared to traditional oral exams, which are typically time-intensive and difficult to administer at scale. By automating both questioning and grading, the system enables efficient assessment of large student populations.

Third, the platform offers strong advantages in academic integrity. Unlike written assessments, real-time oral responses are more difficult to outsource or generate using external tools, reducing the risk of dishonest behavior.

In addition, the solution provides flexibility through configurable questions and adaptive interaction, making it applicable across different subjects and learning contexts. Finally, the system demonstrates high efficiency, with grading completed in seconds rather than minutes.

While the proposed approach introduces new challenges, such as dependence on prompt design and the need for further system refinement, it offers the most comprehensive solution when evaluated against the defined criteria.

For these reasons, the AI-powered oral examination platform was selected as the final solution for this project.

Literature Survey

Introduction

The rapid advancement of Artificial Intelligence (AI), particularly Large Language Models (LLMs), is transforming the landscape of higher education. Recent developments in AI-powered systems have enabled automated grading, personalized feedback, and scalable educational support, significantly reducing instructor workload while enhancing student learning experiences (Deepshikha, 2025; Kasneci et al., 2023). However, these advancements have also introduced new challenges, particularly in the area of academic integrity and assessment reliability.

Traditional assessment methods, especially written and take-home exams, are increasingly vulnerable to misuse due to the accessibility of generative AI tools. As LLMs are capable of producing high-quality academic responses that can bypass conventional plagiarism detection systems, institutions are facing a critical need to redesign evaluation frameworks (Cotton et al., 2024). This shift necessitates the development of more interactive, real-time, and authentic forms of assessment that can accurately measure a student's understanding.

This literature review is conducted to support the proposed development of an AI-powered oral examination platform that leverages a "Council-of-LLMs" framework to evaluate student responses in real time. Unlike traditional research-focused literature reviews, this review aims to justify the feasibility and applicability of a technological solution designed to address a specific problem within higher education assessment systems.

The review is organized into the following thematic sections: Industry context, the Problem being addressed, the Proposed Solution, underlying Technology, relevant Use Cases, and a concluding synthesis of findings. Brief overview of structure and focus of literature review

The Industry

The higher education sector has increasingly adopted AI-driven technologies to enhance both teaching and assessment processes. According to Zawacki-Richter et al. (2019), AI applications in higher education can be broadly categorized into assessment, adaptive learning, intelligent tutoring systems, and predictive analytics. Among these, automated grading systems have gained significant attention due to their potential to improve efficiency and scalability.

Recent studies indicate that AI-powered grading systems can significantly reduce grading time while providing consistent and personalized feedback to students (Deepshikha, 2025). In STEM education, advanced AI models, including transformer-based architectures, have demonstrated the ability to evaluate complex, open-ended responses by analyzing semantic meaning rather than relying solely on keyword matching (Tan et al., 2025). This shift represents a major evolution from traditional rule-based grading systems to more sophisticated, context-aware evaluation methods.

Despite these advancements, the integration of AI into educational assessment also presents challenges. Issues related to bias, transparency, and ethical implementation remain critical concerns (Yan et al., 2024). Furthermore, there is a growing recognition that technological solutions must be aligned with pedagogical goals and involve educators in their design and deployment (Zawacki-Richter et al., 2019). These factors highlight both the opportunities and complexities of applying AI within the education industry. Discussion of relevant industry background and landscape

The Problem

The increasing accessibility of generative AI tools has fundamentally disrupted traditional assessment methods in higher education. Written assignments and take-home exams, once considered reliable indicators of student understanding, are now highly susceptible to AI-generated responses. Research by Cotton et al. (2024) demonstrates that LLMs can produce high-quality academic content that is difficult to distinguish from human work, effectively bypassing existing plagiarism detection systems.

This challenge extends beyond academic dishonesty to a broader issue of assessment validity. Traditional evaluation methods often fail to capture a student's true cognitive processes, as they focus primarily on final outputs rather than reasoning and understanding. As a result, educators are unable to reliably assess whether students genuinely comprehend the material or are simply leveraging AI tools to generate answers.

In addition to concerns about academic integrity, the use of AI in education introduces ethical challenges, including bias, lack of transparency, and accountability (Yan et al., 2024). AI systems may exhibit inconsistencies when evaluating diverse linguistic expressions, accents, or cultural communication styles, potentially leading to unfair outcomes. Moreover, "black box" decision-making processes in AI grading systems can undermine trust if students are unable to understand how their scores are determined.

Given these limitations, there is a clear need for a new form of assessment that is interactive, resistant to AI misuse, and capable of evaluating genuine student understanding. This problem forms the foundation for the proposed solution.

The Proposed Solution

To address these challenges, this project proposes the development of an AI-powered oral examination platform that evaluates student responses in real time using a multi-agent "Council-of-LLMs" framework. Unlike traditional written assessments, this system emphasizes live interaction, requiring students to articulate their understanding verbally, thereby reducing the likelihood of AI-assisted cheating.

The feasibility of this approach is supported by existing research on AI-based grading systems. Studies have shown that LLMs are capable of evaluating open-ended responses with a high degree of accuracy by analyzing underlying concepts rather than surface-level features (Gao et al., 2024; Pecuchova et al., 2025). Furthermore, AI-assisted grading systems have

demonstrated strong correlation with human evaluators, indicating their reliability in educational contexts (Chamberlain et al., 2025).

The proposed system builds upon these capabilities by introducing a real-time, conversational assessment format. By engaging students in dynamic dialogue, the platform can assess not only their final answers but also their reasoning processes, critical thinking skills, and ability to respond to follow-up questions. Additionally, the use of multiple AI agents working collaboratively enhances consistency and reduces the risk of biased or inaccurate evaluations.

Overall, this solution represents a shift toward more authentic and interactive assessment methods, aligning with recommendations from the literature to redesign evaluation frameworks in the age of AI.

The Technology

The proposed solution is enabled by several key technological components, including Large Language Models, natural language processing, multi-agent systems, and fairness-aware AI design.

Large Language Models (LLMs) serve as the core of the system, providing the ability to understand and evaluate complex, open-ended responses. Unlike traditional grading systems that rely on predefined rules or keyword matching, LLMs can interpret semantic meaning and assess the quality of a student’s reasoning (Gao et al., 2024). This capability allows for more nuanced and accurate evaluation of student performance.

In addition to individual model capabilities, the proposed system incorporates a multi-agent architecture, referred to as a “Council-of-LLMs.” In this framework, multiple AI models independently evaluate a student’s response and collectively determine a final assessment. This approach enhances reliability by reducing the impact of individual model biases or errors, leading to more consistent outcomes.

Another critical aspect of the technology is the integration of fairness and bias mitigation strategies. As highlighted by Mehrabi et al. (2021), machine learning systems can be affected by various forms of bias, including those related to language, culture, and data representation. To address these concerns, the proposed system includes mechanisms for monitoring and mitigating bias, ensuring equitable evaluation across diverse student populations.

Transparency and accountability are also essential components of the system. According to Binns (2018), algorithmic decision-making must be accompanied by clear and understandable explanations. In line with this principle, the platform is designed to provide detailed feedback and justification for each assessment, allowing students to understand how their performance was evaluated.

Finally, ethical considerations are integrated into the system design through an “Ethics-by-Design” approach (Holmes et al., 2022). This includes ensuring that the technology supports, rather than replaces, the role of educators and aligns with broader pedagogical objectives.

Use Cases

Several existing applications of AI in education demonstrate the feasibility of the proposed solution. For example, AI-assisted grading systems in medical and engineering education have shown strong alignment with human evaluators, achieving high levels of accuracy in scoring open-ended responses (Chamberlain et al., 2025).

In programming education, automated grading tools have been widely adopted to provide immediate feedback and support student learning at scale (Messer et al., 2024). These systems have proven effective in improving student engagement and learning outcomes by reducing delays in feedback.

Additionally, research on LLM-based grading of conceptual questions in engineering demonstrates that AI systems can evaluate not only correctness but also depth of understanding and reasoning (Gao et al., 2024). These use cases provide strong evidence that AI technologies can be successfully applied to complex assessment scenarios.

Together, these examples support the viability of extending AI-based grading to real-time oral examinations, as proposed in this project.

Conclusion

The literature reviewed in this study highlights both the opportunities and challenges associated with the use of AI in educational assessment. While AI-powered grading systems have demonstrated significant potential in improving efficiency, scalability, and consistency, traditional assessment methods are increasingly inadequate in the face of generative AI technologies.

Most studies agree that there is a pressing need to redesign assessment frameworks to ensure academic integrity and accurately measure student understanding. At the same time, concerns related to bias, transparency, and ethical implementation must be carefully addressed.

The proposed AI-powered oral examination platform aligns with these findings by offering a novel approach that combines real-time interaction, advanced language understanding, and multi-agent evaluation. By addressing both the technical and ethical challenges identified in the literature, this solution represents a promising direction for the future of educational assessment.

Approach and Methodology

Problem Statement and Research Question

The widespread adoption of generative AI has fundamentally undermined the effectiveness of traditional written and take-home assessments. Students can now produce sophisticated written work with minimal understanding, making it increasingly difficult for faculty to distinguish genuine learning from AI-assisted output. While existing AI detection tools and prevention mechanisms exist, they are inherently unreliable and place significant burdens on instructors. In contrast, oral examinations have long been recognized as the gold standard for authentic assessment, as they require direct engagement with the subject matter and cannot be easily circumvented by generative AI.

However, despite their assessment validity, oral examinations at scale present significant logistical and practical challenges. Traditional oral exams are extremely time-consuming, difficult to schedule across large cohorts, and do not scale to typical class sizes. As a result, NYU School of Professional Studies (NYU SPS) faces a critical challenge: how to restore assessment validity in a way that is simultaneously scalable, cost-effective, and practical for faculty.

To address this challenge, the following research question was formulated:

" Can an AI-powered oral examination platform effectively replicate the benefits of traditional oral exams while achieving scalability, consistency, and reliability in academic assessment?"

Proof of Concept Approach

The approach of this project was to develop a proof-of-concept AI-powered oral examination platform that simulates the structure and evaluation process of traditional oral assessments using large language models and voice interaction technologies. The system was designed as a web-based application integrating conversational AI, automated grading, and real-time feedback.

The core architecture consists of three main components: (1) an AI conversational engine responsible for generating exam questions and maintaining dialogue flow, (2) a voice interaction layer that enables text-to-speech (TTS) and speech-to-text (STT) capabilities, and (3) an evaluation module that assesses student responses using a rubric-guided large language model.

The OpenAI GPT-4o-mini model was selected as the primary evaluation engine due to its strong performance in natural language understanding, reasoning, and response evaluation. ElevenLabs was used to generate natural-sounding audio prompts for the oral exam experience. The backend system was implemented using FastAPI, enabling scalable API-based interaction between components.

A key design element of the system is prompt engineering. The evaluation prompt was carefully structured to include explicit grading criteria (e.g., accuracy, completeness, clarity, and

conceptual understanding) and defined score ranges. This rubric-guided approach ensures that the model produces consistent and interpretable scores aligned with academic assessment standards.

The proof of concept focuses on demonstrating feasibility rather than full production deployment. Therefore, exam questions were predefined within the system, and evaluation was conducted under controlled simulated conditions.

Technology Trial Plan

To evaluate the effectiveness of the proposed system, a structured simulation-based technology trial was conducted. Instead of using real student participants, representative student responses were generated to simulate different levels of performance. This controlled setup allows for systematic evaluation of grading behavior, consistency, and differentiation capability.

The trial was designed to test three key aspects of the system:

- The ability to differentiate between high-, medium-, and low-quality responses
- The consistency of grading across repeated evaluations
- The realism of aggregated student performance (mixed response quality scenarios)

The trial environment replicates a simplified oral exam setting where each student responds to five conceptual questions, and each response is evaluated independently by the AI grading system.

Population/Data

A simulated dataset was constructed to represent a small cohort of students with varying levels of performance. Five student profiles were defined:

- High-performing student (consistently strong responses)
- Medium-performing student (moderately accurate responses)
- Low-performing student (consistently weak responses)
- Mixed-performance student (combination of high, medium, and low responses)
- Mixed-performance student with lower overall performance

Each student responded to five predefined questions, resulting in a total of 25 evaluated responses.

The independent variable in this experiment is response quality level (high, medium, low), which reflects the conceptual completeness and clarity of answers. The dependent variable is the AI-generated score assigned by the evaluation model.

Additional derived variables include:

- Average score per student
- Standard deviation of scores (to measure performance variability)

To evaluate grading consistency, a subset of responses (high, medium, and low) was evaluated multiple times under identical conditions.

Procedures

The experimental procedure followed a structured workflow:

1. **Question Setup**
A fixed set of five conceptual questions was defined within the system for each exam session.
2. **Response Simulation**
Representative responses were created for each quality level (high, medium, low), ensuring clear differentiation in accuracy, depth, and clarity.
3. **Evaluation Process**
Each response was submitted to the AI evaluation module, which used a rubric-based prompt to generate a score and feedback.
4. **Data Recording**
Scores for each question were recorded and aggregated to compute average scores and standard deviation for each student.
5. **Consistency Testing**
Selected responses were evaluated multiple times to assess the stability of scoring outputs.
6. **Data Compilation**
All results were organized into structured tables for analysis, including performance distribution and consistency measurements.

Measurements

The following metrics were used to evaluate system performance:

- **Average Score:** Measures overall student performance across questions
- **Score Distribution:** Evaluates differentiation between performance levels
- **Standard Deviation:** Measures consistency of performance within a student's responses
- **Repeated Scoring Variation:** Measures grading consistency across multiple evaluations of the same response

These metrics were selected to align with the project objectives of assessing grading reliability, differentiation capability, and scalability.

Data Collection Methodology

All data was generated and collected within the system through controlled simulation. Scores were produced by the AI evaluation module and recorded programmatically during each interaction.

For consistency evaluation, identical responses were submitted multiple times, and the resulting scores were manually recorded and compared.

The dataset was organized using spreadsheet tools (e.g., Excel) to calculate summary statistics such as averages and standard deviations.

Data Analysis

Data analysis focused on identifying patterns in scoring behavior and validating system performance against the defined objectives.

The analysis included:

- Comparing average scores across student types to evaluate differentiation
- Examining standard deviation values to assess consistency within responses
- Analyzing repeated scoring results to evaluate grading stability
- Observing whether mixed-performance students received intermediate scores reflecting their response distribution

Descriptive statistical methods were used to summarize results, as the primary goal was to demonstrate feasibility rather than conduct inferential statistical testing.

Organizational Change Plan

The adoption of an AI-powered oral examination system presents several organizational challenges. First, faculty members may be resistant to relying on automated grading systems due to concerns about accuracy, transparency, and loss of control over the assessment process. Second, students may initially distrust AI-generated evaluations, particularly if they do not understand how scores are determined.

To address these challenges, a phased adoption strategy is proposed. Instructors will be involved in configuring grading rubrics and reviewing system outputs to ensure alignment with academic standards. Transparency will be emphasized through detailed feedback and clear scoring criteria. Additionally, the system can initially be deployed as a supplementary practice tool rather than a high-stakes assessment mechanism, allowing both instructors and students to build trust gradually.

Training sessions and documentation will be provided to support faculty adoption, while continuous feedback from users will be incorporated into system improvements. This approach

ensures that the platform is introduced in a way that supports, rather than disrupts, existing educational practices.

Results

Data Processing

The dataset used in this study was generated through a controlled simulation of oral examination scenarios. Five student profiles were created to represent different levels of performance: high, medium, low, and two mixed-performance profiles. Each student responded to five predefined conceptual questions.

All responses were evaluated using the AI grading module, and the resulting scores were recorded. The data was organized in a structured format and processed using spreadsheet tools to calculate summary statistics, including average scores and standard deviation for each student.

No additional data cleaning was required, as the dataset was generated under controlled conditions with predefined response quality levels.

Findings

The results demonstrate that the AI-powered oral examination system is capable of effectively differentiating between varying levels of student performance, maintaining grading consistency, and producing realistic evaluation patterns.

Score Differentiation

The system successfully distinguished between high-, medium-, and low-performing students. The high-performing student achieved the highest average score (91/100), while the medium-performing student received an average score of 83/100. The low-performing student consistently received the lowest scores, with an average of 60/100.

These results indicate that the grading mechanism is sensitive to differences in response quality and can assign scores that align with expected academic performance levels.

Table 1. Student Performance Scores Across Questions

	A	B	C	D	E	F	G	H
1	Student	Question 1	Question 2	Question 3	Question 4	Question 5	Avg Score	Std Dev
2	Student 1 (High)	95	90	90	90	90	91	2.23606798
3	Student 2 (Medium)	90	85	80	80	80	83	4.47213595
4	Student 3 (Low)	60	60	60	60	60	60	0
5	Student 4 (Mixed: 2H/2M/1L)	95	90	80	80	60	81	13.4164079
6	Student 5 (Mixed: 1H/2M/2L)	90	85	80	60	60	75	14.1421356

Mixed Performance Behavior

To simulate more realistic student behavior, two mixed-performance profiles were introduced. These students provided a combination of high-, medium-, and low-quality responses across different questions.

The results show that mixed-performance students received intermediate average scores. Specifically, one mixed student achieved an average score of 81/100, while the other achieved

75/100. These scores fall between the high and medium, and medium and low performance levels, respectively.

This demonstrates that the system can accurately aggregate varied response quality into a coherent overall assessment, reflecting realistic fluctuations in student performance.

Standard Deviation Analysis

Standard deviation was calculated to measure the variability of scores across questions for each student.

The high-performing student exhibited a low standard deviation (approximately 2.2), indicating consistent high-quality responses. Similarly, the low-performing student showed no variation (standard deviation = 0), as all responses were uniformly weak.

In contrast, mixed-performance students exhibited significantly higher standard deviation values (above 13), reflecting variation in response quality across questions. The medium-performing student showed moderate variability (standard deviation ≈ 4.5).

These results suggest that the system captures not only average performance but also variability in student responses, providing a more nuanced evaluation of performance consistency.

Grading Consistency (Repeated Evaluation)

To evaluate the reliability of the grading mechanism, selected responses representing high-, medium-, and low-quality performance were evaluated multiple times under identical conditions.

The results show minimal variation across repeated evaluations. The high-quality response received scores of 95, 95, and 90, while the medium-quality and low-quality responses showed no variation, consistently receiving scores of 80 and 60, respectively.

This indicates that the system demonstrates a reasonable level of grading consistency, with only minor fluctuations observed in certain cases. Such variability is expected in AI-based evaluation systems and remains within an acceptable range.

Table 2. Grading Consistency Across Repeated Evaluations

	A	B	C	D	E
1	Response Ty	Question	Attempt 1	Attempt 2	Attempt 3
2	High	Q1	95	95	90
3	Medium	Q3	80	80	80
4	Low	Q5	60	60	60

Summary Statistics

Overall, the results indicate a clear and consistent pattern in scoring behavior:

- High-performing students received the highest scores

- Low-performing students received the lowest scores
- Mixed-performance students received intermediate scores
- Standard deviation values reflected performance consistency
- Repeated evaluations showed stable grading behavior

These findings support the conclusion that the system can replicate key characteristics of human grading, including differentiation, consistency, and responsiveness to varying answer quality.

Figure 1. AI Oral Exam Interface

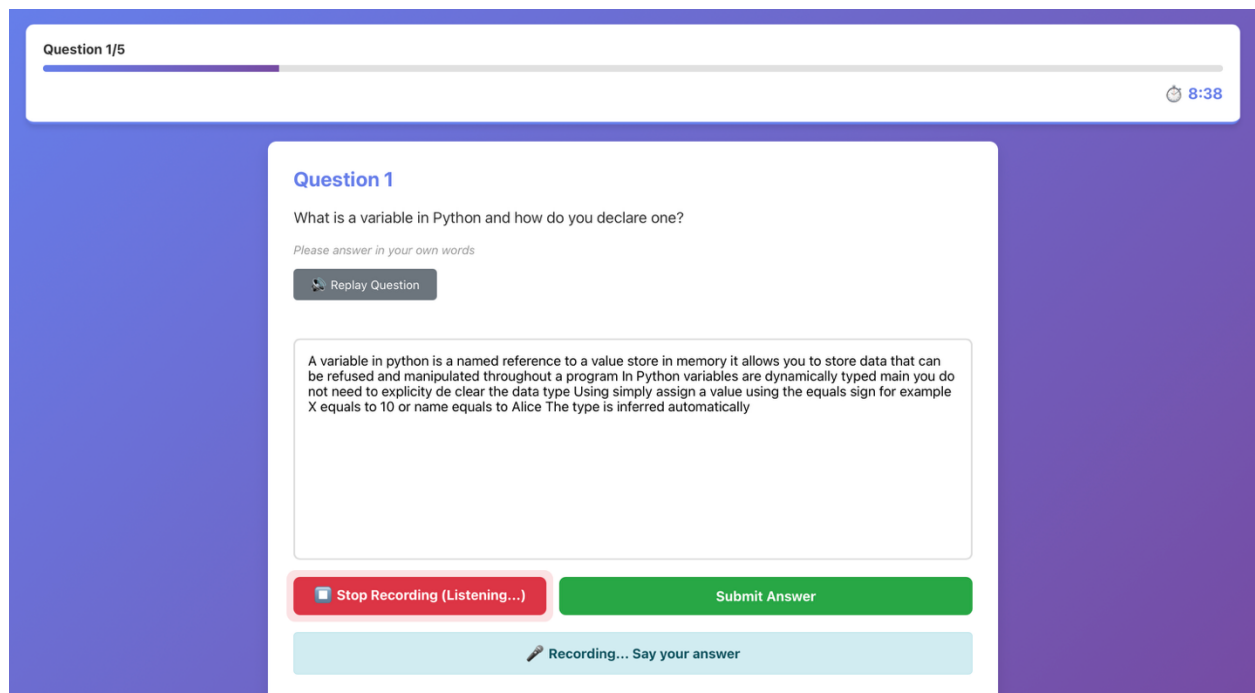


Figure 2. AI-Generated Feedback and Scoring Output

Question Responses

Question 1

90/100

Question:

What is a variable in Python and how do you declare one?

Your Answer:

A variable in Python is a named reference to a value stored in memory. It allows you to store data that can be reused and manipulated throughout a program. In Python, variables are dynamically typed, meaning you do not need to explicitly declare their data type. You simply assign a value using the equals sign, for example: `x = 10` or `name = "Alice"`. The type is inferred automatically.

Feedback:

Great job! You demonstrated a strong understanding of variables in Python and how to declare them. Adding a bit more detail could enhance your response, but overall, it was well done.

Strengths:

The answer accurately defines a variable in Python and explains its purpose clearly. The explanation of dynamic typing and how to declare a variable is correct and includes appropriate examples.

Areas for Improvement:

The answer could further elaborate on the implications of dynamic typing or provide more examples of different data types.

Outcomes

The results of the simulation demonstrate that the AI-powered oral examination platform meets its core functional objectives. The system is capable of evaluating open-ended responses, distinguishing between levels of understanding, and generating consistent scores across repeated evaluations.

In addition, the system produces realistic assessment patterns, particularly in mixed-performance scenarios, where student performance varies across questions. This suggests that the platform can effectively model real-world academic evaluation conditions.

Implications

The findings have important implications for the use of AI in academic assessment. First, the system demonstrates the potential to automate oral exams while maintaining key qualities of traditional evaluation, including depth of assessment and performance differentiation.

Second, the ability to produce consistent and scalable grading suggests that such systems can significantly reduce instructor workload, particularly in large classes or online environments.

Finally, the results support the feasibility of using AI-powered oral examinations as a complementary or alternative assessment method in higher education, particularly in contexts where academic integrity is a concern.

Summary

In summary, the experimental results confirm that the proposed AI-powered oral examination platform can effectively evaluate student responses, maintain grading consistency, and produce realistic performance distributions. The system demonstrates strong potential as a

scalable and reliable assessment solution, while also highlighting areas for further refinement in future development.

Repository of Data Sets and Code

<https://github.com/danxia2002/oral-exam-platform.git>

Issues Encountered

While developing the AI-powered oral examination platform, several technical and project-related challenges were encountered. Although none of these issues critically impacted the overall project timeline or deliverables, they provided valuable insights into system design, AI behavior, and risk management. The following sections describe the key issues and the strategies used to resolve them.

Prompt Design and Grading Inconsistency

One of the most significant challenges encountered during the project was the inconsistency in AI-generated grading results. Initially, the evaluation prompt used to assess student responses did not include a clearly defined grading rubric. As a result, the large language model tended to assign overly conservative scores, even for high-quality answers. In some cases, strong responses were scored in the range of 60–70, which did not align with expected academic standards.

This issue was directly related to Risk 3 identified in the risk management plan: “*AI model generates inaccurate or inconsistent evaluation results.*”

To address this challenge, prompt engineering techniques were applied to redesign the evaluation prompt. A structured scoring rubric was introduced, explicitly defining score ranges (e.g., 90–100 for excellent responses, 80–89 for good responses, etc.) and emphasizing key evaluation criteria such as accuracy, completeness, clarity, and conceptual understanding. Additionally, instructions were added to encourage fair and non-overly strict grading.

Following this modification, the grading outputs became more aligned with expected performance levels, and the system demonstrated improved differentiation between high-, medium-, and low-quality responses. This experience highlights the critical importance of prompt design in controlling LLM behavior and ensuring reliable system outputs.

Variability in Repeated Evaluations

Another issue observed during system testing was the variability in scores when the same response was evaluated multiple times. Although the differences were relatively small, slight fluctuations (e.g., a score of 95 versus 90 for the same response) were noted.

This issue is inherent to probabilistic AI models and was also anticipated in the risk management plan under evaluation reliability concerns.

To mitigate this issue, the model’s temperature parameter was adjusted to a lower value, reducing randomness in output generation and improving consistency. In addition, a repeated evaluation experiment was conducted to measure the extent of variability. The results showed that score differences remained within a small and acceptable range, indicating that the system maintains reasonable grading stability.

Rather than eliminating variability entirely, the project addressed it through measurement and validation, demonstrating that the system’s behavior is consistent enough for practical use.

Integration Challenges with External APIs

The system relies on multiple external APIs, including OpenAI for language processing and ElevenLabs for text-to-speech functionality. During development, intermittent issues were encountered related to API response delays and occasional failures in audio generation.

This aligns with Risk #7 in the risk management plan: *“Internet/API dependency causing delays or failures.”*

To address this, fallback mechanisms were implemented. For example, when audio generation failed, the system continued to function using text-based output. This ensured that the core exam flow was not interrupted. Additionally, error handling was incorporated into the backend to capture and manage API failures gracefully.

These measures allowed the system to remain stable and functional even under imperfect external conditions.

Time Constraints and Scope Management

Given the limited timeframe of a single academic semester, managing project scope was a critical challenge. There was an initial tendency to expand the system to include advanced features such as adaptive questioning, multi-agent grading (“Council-of-LLMs”), and full user interface enhancements.

This issue corresponds to Risk #9: *“Scope creep (adding too many features beyond initial plan).”*

To mitigate this risk, a strict focus on the Minimum Viable Product (MVP) was maintained. Priority was given to core functionalities, including:

- Question delivery
- Response evaluation
- Score generation
- Basic feedback

More advanced features were documented as future work rather than being implemented within the current project scope. This approach ensured that the system remained functional, testable, and aligned with project deadlines.

Simulation-Based Data Limitations

Another limitation encountered was the use of simulated data instead of real student responses. While simulation allowed for controlled testing of system behavior, it does not fully capture the complexity and variability of real-world student interactions.

This was not a direct failure but rather a constraint imposed by project scope and available resources.

To address this, the simulation was carefully designed to include multiple performance levels (high, medium, low, and mixed) and realistic response variations. Additionally, the

methodology explicitly acknowledges the use of simulated data, ensuring transparency in the evaluation process.

Future work will involve conducting real-world pilot studies with actual students to validate system performance in practical settings.

Risk management plan

A comprehensive risk management plan was developed at the beginning of the project to identify potential challenges and define mitigation strategies. The key risks included development delays, integration issues, AI evaluation inaccuracies, and scope creep.

Several of these risks materialized during the project, particularly those related to AI grading behavior and API dependencies. However, the predefined mitigation strategies proved effective in addressing these issues.

For example, prompt refinement and controlled experimentation were used to manage AI evaluation risks, while fallback mechanisms and error handling addressed integration-related challenges. Scope management strategies ensured that the project remained focused and achievable within the given timeframe.

Overall, the risk management plan played a critical role in guiding decision-making and ensuring that issues were resolved efficiently without compromising the quality of the final deliverable.

Summary

In summary, the project encountered several technical and operational challenges, primarily related to AI behavior, system integration, and project scope. Through a combination of prompt engineering, system design adjustments, and risk management strategies, these issues were successfully addressed.

These experiences not only improved the robustness of the system but also provided valuable insights into the practical challenges of implementing AI-powered solutions in real-world applications.

Lessons Learned

The successful completion of this project provided valuable learning experiences across technical, analytical, and project management domains. Throughout the planning, development, and evaluation phases, several key lessons were gained that contribute to both professional growth and a deeper understanding of AI-driven systems in real-world applications.

Understanding the Importance of Prompt Engineering

One of the most significant lessons learned from this project is the critical role of prompt engineering in controlling the behavior and output quality of large language models. Initially, the grading system produced inconsistent and overly conservative scores due to the lack of clearly defined evaluation criteria. This demonstrated that LLMs, while powerful, do not inherently follow academic grading standards without explicit guidance.

Through iterative refinement, a structured rubric-based prompt was developed, which significantly improved scoring accuracy, consistency, and interpretability. This experience highlighted that effective use of AI systems requires not only technical integration but also careful design of instructions and constraints. Prompt engineering emerged as a core skill in ensuring reliable and meaningful AI outputs.

Designing Experiments for AI Evaluation

Another important lesson was learning how to design evaluation methodologies for AI systems. Unlike traditional software systems, AI-driven applications require validation through carefully structured experiments rather than deterministic testing.

In this project, a simulation-based approach was used to evaluate system performance. This included designing datasets that represent different levels of student performance (high, medium, low, and mixed), as well as conducting repeated evaluations to measure grading consistency. Through this process, it became clear that defining appropriate metrics—such as average scores, standard deviation, and consistency variation—is essential for demonstrating system effectiveness.

This experience strengthened the ability to translate abstract research questions into measurable evaluation frameworks.

Balancing System Complexity and Practical Constraints

During the development process, there was an initial inclination to implement advanced features such as multi-agent grading (Council-of-LLMs) and adaptive questioning mechanisms. However, time and resource constraints required prioritizing a functional and testable minimum viable product (MVP).

This led to an important lesson in scope management: focusing on core functionality is often more valuable than attempting to implement overly complex features within limited timeframes. By concentrating on essential components—such as question delivery, response

evaluation, and feedback generation—the project was able to deliver a complete and demonstrable system.

This experience reinforced the importance of aligning technical ambition with practical feasibility.

Integrating Multiple Technologies into a Cohesive System

The project required the integration of multiple technologies, including large language models (OpenAI API), text-to-speech systems (ElevenLabs), and a backend framework (FastAPI). Managing these components and ensuring seamless interaction between them provided valuable experience in system architecture and API-based development.

One key lesson was the importance of designing modular systems, where individual components can function independently and be tested separately. This approach made it easier to identify and resolve integration issues, particularly when dealing with external API dependencies.

Additionally, handling API failures and implementing fallback mechanisms emphasized the need for robust system design in real-world applications.

Recognizing the Limitations of AI Systems

Another critical insight gained from this project is an understanding of the limitations of AI-based evaluation systems. While the platform demonstrated strong capabilities in differentiating performance levels and providing consistent feedback, it also exhibited minor variability in repeated scoring and dependence on prompt design.

These limitations highlight that AI systems should not be treated as fully autonomous decision-makers but rather as tools that support and augment human judgment. Recognizing these constraints is essential for responsible deployment of AI in educational and professional settings.

Applying Risk Management in Practice

The project also provided practical experience in risk identification and mitigation. Several anticipated risks—such as grading inconsistency, API dependency, and scope creep—were encountered during development.

By applying predefined mitigation strategies, such as prompt refinement, system fallback mechanisms, and strict scope control, these challenges were effectively managed without significantly impacting the project timeline or deliverables.

This demonstrated the value of proactive risk planning and reinforced the importance of structured project management practices in ensuring successful outcomes.

Summary

Overall, this project provided a comprehensive learning experience that combined technical development, experimental design, and project management. Key lessons included the

importance of prompt engineering, the need for structured evaluation methodologies, the value of scope management, and the practical challenges of integrating AI systems into real-world applications.

These insights not only contributed to the successful completion of the project but also established a strong foundation for future work in AI-driven system design and implementation.

Conclusion and Further Work

Conclusions

Overall Outcomes

This project set out to address the research question: *Can an AI-powered oral examination platform effectively replicate the benefits of traditional oral exams while achieving scalability, consistency, and reliability in academic assessment?*

Based on the development and evaluation of the proof-of-concept system, the results indicate that the proposed solution successfully addresses this question. The platform demonstrated the ability to simulate an interactive oral examination process, evaluate open-ended responses, and provide structured feedback in a scalable and efficient manner.

Key Findings

Several key findings emerged from the experimental evaluation:

First, the system effectively differentiates between varying levels of student performance. High-quality responses consistently received higher scores, while low-quality responses were assigned lower scores. Mixed-performance scenarios produced intermediate results, reflecting realistic variations in student ability.

Second, the system demonstrated a reasonable level of grading consistency. Repeated evaluations of the same responses showed minimal variation, indicating that the rubric-guided prompt design contributed to stable scoring behavior.

Third, the system captured not only average performance but also variability in responses. Standard deviation analysis showed that consistent performers had low variability, while mixed-performance students exhibited higher variability, reflecting realistic assessment patterns.

Finally, the system significantly improves efficiency. Responses were evaluated within seconds, compared to several minutes required for manual grading, highlighting strong potential for scalability in large classroom settings.

Viability Assessment

The proof of concept can be considered a success based on both functional performance and evaluation outcomes. The system successfully demonstrates the feasibility of automating oral examinations using large language models and voice-based interaction.

Evidence supporting this conclusion includes:

- Successful execution of end-to-end oral exam simulation
- Clear differentiation of student performance levels
- Stable grading results across repeated evaluations
- Efficient processing and real-time feedback generation

While the system is not yet production-ready, it provides a strong foundation for further development and validation.

Implications

Theoretical Implications

This project contributes to the growing field of AI-driven assessment by demonstrating that large language models can be used to evaluate not only written responses but also simulated oral interactions. It supports the argument that AI systems can approximate key aspects of human evaluation, including reasoning assessment and performance differentiation.

Additionally, the use of structured prompt engineering as a mechanism for enforcing evaluation standards highlights the importance of human-in-the-loop design in AI systems. This reinforces the idea that AI effectiveness depends not only on model capability but also on how it is guided and constrained.

Practical Implications

From a practical perspective, the system offers a scalable alternative to traditional oral examinations. It enables institutions to assess larger numbers of students without proportional increases in instructor workload.

The platform also addresses growing concerns around academic integrity. By requiring real-time verbal responses, the system reduces the likelihood of AI-assisted cheating and promotes authentic demonstration of understanding.

Furthermore, the system can serve as a valuable practice tool for students, providing immediate feedback and helping them improve their ability to articulate concepts in an oral format.

Limitations

Constraints

Despite its promising results, the project has several limitations. The most significant constraint is the use of simulated data rather than real student interactions. While simulation allows for controlled evaluation, it does not fully capture the complexity of real-world responses, including variations in language, communication style, and domain knowledge.

Additionally, the prototype relies on predefined exam questions embedded within the system, limiting flexibility for instructors. The system also depends on external APIs, which may introduce latency or reliability issues.

Validity and Bias Considerations

The system's evaluation process is inherently dependent on the behavior of large language models, which may introduce bias or inconsistencies. Factors such as language style, phrasing, or cultural differences could potentially influence scoring outcomes.

Moreover, grading performance is sensitive to prompt design. Although structured prompts improved consistency, the system may still produce different results under alternative prompt configurations. This highlights the need for further validation and calibration in diverse settings.

Further Work

Next Steps

Several immediate improvements can be made to enhance the system:

- Implement dynamic question generation instead of predefined question sets
- Refine grading prompts and introduce multi-agent evaluation (Council-of-LLMs) to improve reliability
- Conduct pilot studies with real students to validate system performance in practical settings
- Enhance the user interface to improve usability and user experience

In addition, integrating speech-to-text processing more robustly and optimizing response latency will further improve system performance.

Long-term Directions

In the long term, the platform can be expanded into a fully deployable academic assessment system. Potential future developments include:

- Integration with learning management systems (LMS) such as Canvas or Blackboard
- Development of adaptive questioning algorithms that adjust difficulty based on student performance
- Implementation of fairness-aware evaluation mechanisms to mitigate bias
- Expansion to support multiple disciplines and more complex assessment formats

The concept can also be extended beyond education into business training, professional certification, and interview simulation applications.

Repository of Data Sets and Code

The source code for the AI-powered oral examination platform, along with the simulated datasets and analysis files used in this study, are available at the following repository:

<https://github.com/danxia2002/oral-exam-platform.git>

The repository includes:

- Backend application code (FastAPI)
- Prompt design and evaluation logic
- Simulated datasets used for analysis

- Spreadsheet files containing score calculations and statistical summaries
- A copy of this final report

Closing Summary

In conclusion, this project demonstrates the feasibility of using AI-powered systems to replicate key aspects of oral academic assessment while achieving scalability and efficiency. The results show that large language models, when guided by structured prompts and integrated into a well-designed system, can provide consistent, meaningful, and realistic evaluation of student performance.

Although limitations remain, the project provides a strong foundation for future research and development in AI-driven assessment systems. As educational institutions continue to adapt to the challenges introduced by generative AI, solutions such as the one proposed in this project offer a promising direction for maintaining assessment integrity while enhancing learning experiences.

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Appendix A - Project Acceptance Document

Sponsor's Project Acceptance Document

This document is the means by which your project sponsor formally agrees that your project has been satisfactorily completed and that it meets the project goal and objectives that were set at the onset of the project. It is therefore important that you describe the goal, objectives, and related metrics in the appropriate section below. The "PLAN" section is to be prepared at the beginning of the Your sponsor should provide input and sign where indicated. The signed document will also be a required section in your Project Final Report. This document is a template whose sections may be expanded as necessary.

PLAN

(To be filled out and signed at the **start** of the project)

Project Name: AI-Powered Oral Exam Platform for Scalable and Valid Academic Assessment

Student Name: Danxia Hong

Sponsoring Organization: NYU School of Professional Studies (NYU SPS)

Project Sponsor Name and Title: Andres Fortino, Clinical Associate Professor at NYU School of Professional Studies

Project Sponsor Contact Information (email): agf249@nyu.edu

Planned Start Date: January 2026

Planned End Date: April 2026

PROJECT PLAN

At the project start, show the project goal, the project objectives, and related metrics to be used to show successful project completion. The sponsor should sign each objective and the goal with initials to indicate agreement. **Please use PDF version for signatures**

Project Goal

The goal of this project is to design and develop a scalable, AI-powered oral examination platform that enables faculty to assess genuine student understanding through voice-based interaction. The platform will allow instructors to configure oral exams without custom coding, tailoring questions to course objectives and individual student context. By leveraging voice AI and multi-agent workflows, the system will conduct personalized oral assessments at scale. A council-of-LLMs grading framework will be implemented to ensure consistent, transparent, and evidence-based evaluation. The project aims to deliver a deployable prototype that can be used across multiple courses at NYU SPS within a single academic semester.

Objective #1

Define the scope, requirements, and system architecture for the AI-powered oral exam platform, including exam configuration, student context inputs, grading workflows, and compliance considerations.

Deliverable: Platform requirements and system design document

Metric: Sponsor approval of requirements and architecture by mid-semester

Objective #2

Develop a working prototype of the AI-powered oral exam platform capable of conducting end-to-end voice-based oral exams with configurable topics and automated grading.

Deliverable: Working MVP of the oral exam platform

Metric: Successful live demonstration of a complete oral exam workflow to Professor Fortino

Objective #3

Pilot the platform using representative course scenarios to validate grading consistency, assessment validity, and student experience compared to traditional written assessments.

Deliverable: Pilot evaluation and validation report

Metric: Sponsor acceptance of validation results and grading reliability

Objective #4

Finalize the platform for deployment by delivering documentation, configuration guidelines, analytics dashboards, and instructor support materials.

Deliverable: Deployment-ready platform package, including a documented GitHub repository and final project report

Metric: Sponsor sign-off and project completion by end of semester

I agree with the above-planned project goal, project objectives, and related metrics.



Project Sponsor Signature

02/12/2026

Date

Appendix B - Project Sponsor

New York University MS in Management and Analytics Applied Technical Project Project Sponsor Agreement

Goals of the Program

For Participating Organizations

- Begin a relationship with New York University
- Receive help from a highly trained NYU graduate student
- Provide a practical experience opportunity for NYU graduate students
- Receive assistance at no cost

For NYU Graduate Students

- Manage and implement a meaningful project aligned with their professional and educational goals
- Hands-on experience interacting with a start-up or operational small business or organization
- Earn credit toward completion of a graduate degree by conducting an unpaid Applied Project under the mentorship of an NYU-SCPS professor.

Project Sponsor and Student Responsibilities

- Student prepares project planning documents
- Sponsor reviews and approves the student's project plan
- Student submits project plan to faculty supervisors for approval
- Student conducts project according to plan
- At predetermined milestones, the sponsor reviews and approves the status reports submitted by student
- Status reports reviewed and evaluated by faculty supervisors to assure student effort and project meet course requirements
- Project sponsor and student participate in periodic project reviews with NYU
- At project completion project sponsor completes evaluation forms
- Student prepares final report

Project Selection Process

- The Faculty reviews proposed projects
- Projects are:
 - Relevant to MS degree course content
 - Significant to the participating organization
 - Substantial in terms of duration and scope
 - Challenging to the student
 - Capable of being measured against predetermined goals

The MS in Management and Systems

Concentrations in:

- Business Analytics
- Risk Analytics
- Business Informatics
- Business Research

Typical Participating Student Profile

- Students selected to participate in this program meet stringent criteria
- Have completed all coursework
- High achievers with the highest level GPAs and strong academic credentials
- 2-10 years of business experience
- Highly motivated for success

Sponsor and Project Information

Type of Organization	☐ For Profit ☒ Not for Profit				
Name of Organization	NYU School of Professional Studies (NYU SPS)				
Address	7 East 12th Street				
City	New York	State	NY	Zip	10003
Project Sponsor	First Name	Andres	Last Name	Fortino	
Title	Clinical Associate Professor at NYU School of Professional Studies				
Phone	(845) 242-7614				
Email	agf249@nyu.edu				
Web Site	https://www.sps.nyu.edu				
Type of Business	Higher Education				

Student Name	Danxia Hong
Project Title	AI-Powered Oral Exam Platform for Scalable and Valid Academic Assessment

Description of Project	• Design, build, and validate a web-based AI-powered oral exam platform for scalable academic assessment. • Develop an instructor-facing configuration interface for defining assessment topics, exam structure, grading criteria, and student context. • Implement a student-facing voice-based oral exam experience using conversational AI. • Build an analytics dashboard to support evaluation of assessment outcomes, grading consistency, and instructional insights. • Use a multi-agent workflow architecture to manage authentication, topic-based questioning, and personalized project evaluation. • Implement a council-of-LLMs grading framework with deliberation and synthesis to generate structured, evidence-based feedback. • Deliver a working prototype, comprehensive documentation, and a validation study comparing oral assessment outcomes to traditional written methods.
Estimated Hours of Student Participation	300–350 hours (excluding meetings/travel)

Anticipated Results	• A functional, generalizable AI-powered oral examination platform capable of conducting scalable, personalized voice-based assessments. • Evidence that AI-mediated oral exams improve assessment validity compared to traditional written or take-home evaluations in the presence of generative AI. • Identification of key instructional and assessment gaps revealed through systematic oral evaluation and analytics. • Demonstration that a configurable, low-cost AI platform can significantly reduce faculty time and logistical burden while improving feedback quality. • A replicable assessment framework and deployment model that can be adapted across multiple courses and programs within NYU SPS.
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Knowledge and expertise student will need to be able to complete the project • Multi-agent AI architectures and orchestration • Real-time audio processing and speech interfaces • Cloud deployment and scalable system design • LLM evaluation and automated grading frameworks
--

Will the project sponsor be available for periodic meetings with NYU to review progress, address questions and concerns with the professor supervising the program? <i>This is a requirement for the program</i>	☒ Yes ☐ No
Describe the form and frequency of supervision of the student by the Project Sponsor. The sponsor will supervise the project through weekly progress meetings and ongoing email communication. During these meetings, the sponsor will review milestones, provide feedback on system design and implementation decisions, and approve key deliverables. Additional ad-hoc consultations will be scheduled as needed. The sponsor will also review monthly status reports and the final project deliverables to ensure alignment with academic and instructional objectives.	

Sponsor Agreement

Students are interns, not professional consultants. NYU is not responsible for the outcomes of projects undertaken by students. Work is on a best-efforts basis; no guarantees or warranties are expressed or implied. Organization is responsible for evaluating work presented, determining its value and whether to use it or not. Some projects may require on-going management or even re-work by the Organization after the student completes their Applied Project.

Please note that in order to post an unpaid position, the internship must encompass all 6 components below:

1. The internship, even though it includes actual operation of the facilities of the employer, is similar to training which would be given in an educational environment;
2. The internship experience is for the benefit of the intern;
3. The intern does not displace regular employees, but works under close supervision of existing staff;
4. The employer that provides the training derives no immediate advantage from the activities of the intern; and on occasion its operations may actually be impeded;
5. The intern is not necessarily entitled to a job at the conclusion of the internship; and
6. The employer and the intern understand that the intern is not entitled to wages for the time spent in the internship.

I have read and agree with the information shown in the Terms and Conditions for employers contained on the following web page(s): <http://www.nyu.edu/life/resources-and-services/career-development/employers/post-a-job/terms-and-conditions.html>

Please complete and sign this form in the space provided below and return to the course professor via the student who will upload the document to the course drop-box. For any questions, please email the professor: Prof. Andres Fortino agf249@nyu.edu.

I agree to all of the above

Participating Organization NYU School of Professional Studies Date 02/12/2026

By (signature): 
Project Sponsor

Printed Name: Andres Fortino

Title: Clinical Associate Professor at NYU School of Professional Studies

Student Agreement

Students who are planning to conduct an unpaid Applied Project must read and agree to the "Important Considerations Before Accepting a Job or Internship" contained on the following web page(s): <http://www.nyu.edu/life/resources-and-services/career-development/find-a-job-or-internship/important-considerations-before-accepting-a-job-or-internship.html>.

I agree to all of the above

Student Name (Print) Danxia Hong Date 02/12/2026

Signature: 

Appendix C – Functional Requirements Specifications

Project Goal

The goal of this project is to design and develop a scalable, AI-powered oral examination platform that enables faculty to assess genuine student understanding through voice-based interaction. The platform will allow instructors to configure oral exams without custom coding, tailoring questions to course objectives and individual student context. By leveraging voice AI and multi-agent workflows, the system will conduct personalized oral assessments at scale. A council-of-LLMs grading framework will be implemented to ensure consistent, transparent, and evidence-based evaluation. The project aims to deliver a deployable prototype that can be used across multiple courses at NYU SPS within a single academic semester.

Project Objectives

1. Define the scope, requirements, and system architecture for the AI-powered oral exam platform, including exam configuration, student context inputs, grading workflows, and compliance considerations.

Deliverable: Platform requirements and system design document

Metric: Sponsor approval of requirements and architecture by mid-semester

2. Develop a working prototype of the AI-powered oral exam platform capable of conducting end-to-end voice-based oral exams with configurable topics and automated grading.

Deliverable: Working MVP of the oral exam platform

Metric: Successful live demonstration of a complete oral exam workflow to Professor Fortino

3. Pilot the platform using representative course scenarios to validate grading consistency, assessment validity, and student experience compared to traditional written assessments.

Deliverable: Pilot evaluation and validation report

Metric: Sponsor acceptance of validation results and grading reliability

4. Finalize the platform for deployment by delivering documentation, configuration guidelines, analytics dashboards, and instructor support materials.

Deliverable: Deployment-ready platform package, including a documented GitHub repository and final project report

Metric: Sponsor sign-off and project completion by end of semester

Requirements Specifications

Based on the approved project proposal, the following functional requirements are defined for the AI-powered oral examination prototype:

1. Purpose and Scope

The system will function as an AI-powered oral assessment platform designed to simulate academic oral examinations.

The primary users are graduate-level students and instructors.

The platform will:

- Conduct structured oral exam simulations
- Evaluate responses using LLM-based assessment criteria
- Provide structured performance feedback
- Support instructional alignment with course learning objectives

The prototype will focus on assessment simulation rather than full LMS replacement.

2. User Interaction

- The primary mode of interaction will be voice-based input and output, simulating a real-time oral examination environment.
- Students will respond verbally through microphone input.
- The system will transcribe speech to text for processing and grading purposes.
- The platform will generate spoken prompts using text-to-speech (TTS) technology.
- The system will support natural conversational pacing with minimal response latency.

3. Response Generation

The system will utilize large language models (LLMs) to:

- Interpret student responses
- Maintain conversational context
- Generate adaptive follow-up questions
- Provide rubric-aligned evaluation feedback

The system will maintain conversational memory within a session. Feedback will be structured according to predefined academic criteria (e.g., clarity, analytical depth, theoretical application).

4. Learning and Adaptation

- The prototype will allow configurable grading rubrics.
- The system will log anonymized interaction data for evaluation purposes.
- Adaptive questioning logic will adjust difficulty level based on user responses.

5. Integration with Other Systems

- API integration with OpenAI or similar LLM providers.
- Optional integration with:
 - Course management platforms (future expansion)
 - Authentication system (basic login for prototype)

6. Performance Requirements

- Response time: under 5–8 seconds per generated question.
- System must handle multiple test sessions sequentially.
- Feedback accuracy must align with instructor-defined rubric criteria.
- Stable session continuity without conversation loss.

7. User Interface Design

- Web-based interface.
- Minimalist academic design.
- Clear separation of:
 - Question prompt area
 - Voice response interface
 - Feedback summary panel
- Session progress indicator.

8. Development Platform

The prototype will be developed as a web-based application using:

- Frontend: React or similar framework
- Backend: Python (Flask/FastAPI)
- LLM API integration
- Cloud hosting environment (AWS or equivalent)
- Version control via Git

Agile methodology will be used for iterative development and sponsor feedback.

Prototype Description – Summary

The prototype is a web-based AI oral examination simulator that replicates structured academic oral assessments.

It dynamically generates questions, evaluates student responses using rubric-aligned criteria, and provides structured feedback.

The system demonstrates the feasibility of AI-assisted oral assessment in graduate-level instructional settings.

Prototype Description - Detailed

The proposed AI-powered oral examination platform consists of the following core components:

1. User Authentication Module

The system includes a secure login mechanism that allows authorized users to access the platform. This ensures controlled participation and session tracking within the academic environment.

2. Examination Configuration Module

The platform enables the selection of predefined examination topics aligned with course learning objectives. Assessment criteria and evaluation rubrics are configurable to ensure alignment with instructional standards.

3. AI Conversational Engine

The system integrates a large language model (LLM) to generate structured oral examination prompts. The conversational engine maintains contextual continuity throughout the session and dynamically generates follow-up questions based on the student's responses.

4. Adaptive Questioning Mechanism

The platform adjusts the level of analytical complexity depending on the depth, clarity, and completeness of the student's answers. When responses lack specificity, the system generates clarifying prompts. When responses demonstrate strong conceptual understanding, the system increases question difficulty.

5. Evaluation and Feedback Module

Student responses are analyzed according to predefined rubric categories, including conceptual clarity, analytical depth, application of theory, and communication effectiveness. The system generates structured feedback aligned with these criteria.

6. Feedback Summary Interface

At the conclusion of the session, the platform provides a consolidated performance report. This includes qualitative commentary, strengths, areas for improvement, and targeted recommendations to support further preparation.

7. Technical Architecture

The prototype is designed as a web-based application integrating frontend interface components, backend processing logic, and external LLM API services. The architecture supports scalability and potential future integration with learning management systems.

Use Case

Persona: David Chen, a 27-year-old graduate student in the Management and Systems program at NYU SPS. He is preparing for his final oral examination, which requires him to articulate theoretical frameworks, apply them to real-world business cases, and respond to follow-up analytical questions under time pressure. Although David understands the course materials, he struggles with organizing his thoughts clearly in a live oral exam setting.

Anecdote:

David logs into the AI Oral Exam Simulator on a Thursday evening, one week before his scheduled final oral assessment. He selects the topic “Systems Thinking in Organizational Strategy” from the available exam modules.

The AI examiner begins the session with an open-ended question: “Explain how systems thinking influences long-term organizational strategy.”

David responds verbally through the microphone interface, outlining the core principles but omitting a concrete example. The system transcribes his response in real time for evaluation purposes. Within seconds, the AI examiner generates a follow-up question designed to probe deeper into applied understanding: “Can you provide a real-world example where systems thinking improved strategic decision-making?”

David realizes he needs to connect theory with practice. He revises his answer and continues the simulated examination.

As the session progresses, the system adapts its questioning style based on his responses. When David provides a vague explanation, the system asks clarifying questions. When he demonstrates strong conceptual understanding, the system increases the analytical complexity of the prompts.

At the conclusion of the session, the system generates a structured performance summary:

- Conceptual Clarity: Strong
- Analytical Depth: Moderate
- Real-World Application: Needs Improvement
- Communication Structure: Good, but responses could be more concise

The platform also provides targeted improvement suggestions and recommended practice areas.

By the end of the session, David feels more confident and aware of his strengths and weaknesses. Instead of passively reviewing notes, he has actively practiced responding under simulated exam conditions. He schedules another session later in the week to continue refining his performance before the official oral exam.

Documentation of LLM Prompts (if used)

Portions of this memo were generated with the assistance of a large language model (ChatGPT). The following prompts were used to support drafting:

“Generate a functional requirements specification for an AI-powered oral examination chatbot platform designed for graduate-level assessment, based on Assignment 1B: Final Project Proposal.”

“Write a formal academic use case in narrative format for an AI oral exam simulation system.”

“Revise the prototype description into a professional, structured academic tone suitable for a capstone memo.”

All generated content was reviewed, edited, and adapted to align with the specific objectives and scope of this project.

Appendix D - Project Plan

Project Tasks Outline

Two Level WBS:

Based on the provided project proposal and Functional Requirements Specification (FRS) for the “AI-Powered Oral Exam Platform for Scalable and Valid Academic Assessment” project, I have created the following Work Breakdown Structure (WBS) to three levels of definition with corresponding milestones. The project timeline spans from January 29, 2026 to April 30, 2026.

1. Project Planning (Start: Jan 29, 2026 – End: Feb 26, 2026)

- 1.1 Define project scope and objectives
- 1.2 Finalize functional requirements with sponsor
 - ✧ Milestone: Sponsor approval of project scope and requirements (Feb 14, 2026)
- 1.3 Develop project schedule and WBS

2. Analytical & Evaluation Phase (Start: Feb 27, 2026 – End: Mar 7, 2026)

- 2.1 Conduct Task-Technology Fit (TTF) analysis
 - 2.2 Conduct ECTM evaluation
 - 2.3 Assess governance and compliance risks
- ✧ Milestone: Completion of analytical validation (Mar 7, 2026)

3. System Design Phase (Start: Mar 8, 2026 – End: Mar 26, 2026)

- 3.1 Define system architecture
- 3.2 Design user interaction workflow (voice and text input)
- 3.3 Design feedback generation logic

- ✧ Milestone: Approval of system design framework (Mar 26, 2026)

4. Prototype Planning & Testing Framework (Start: Mar 27, 2026 – End: Apr 16, 2026)

- 4.1 Define pilot testing framework
- 4.2 Develop evaluation metrics
- 4.3 Design comparison study model

- ✧ Milestone: Pilot framework finalized (Apr 16, 2026)

5. Final Review and Documentation (Start: Apr 17, 2026 – End: Apr 30, 2026)

- 5.1 Compile final documentation
- 5.2 Conduct final sponsor review
- 5.3 Submit final project deliverables

- ✧ Milestone: Project completion and submission (Apr 30, 2026)

Three Level WBS

To provide a more detailed three-level Work Breakdown Structure (WBS), the second-level tasks are expanded below. Dates are specified for second-level tasks, while third-level tasks are assumed to fall within their respective parent task timelines.

1. Project Planning (Jan 29, 2026 – Feb 26, 2026)

- **1.1 Define project scope and objectives (Jan 29 – Feb 10)**

- ✧ 1.1.1 Identify core academic problem

- ✧ 1.1.2 Define system objectives
 - ✧ 1.1.3 Establish project constraints
- **1.2 Finalize functional requirements with sponsor (Feb 11 – Feb 17)**
 - ✧ 1.2.1 Gather system input/output specifications
 - ✧ 1.2.2 Review functional requirement categories
 - ✧ 1.2.3 Obtain sponsor approval
- **1.3 Develop project schedule and WBS (Feb 18 – Feb 26)**
 - ✧ 1.3.1 Identify milestones
 - ✧ 1.3.2 Structure WBS hierarchy
 - ✧ 1.3.3 Allocate timeline to phases

2. Analytical & Evaluation Phase (Feb 27, 2026 – Mar 7, 2026)

- **2.1 Conduct Task-Technology Fit (TTF) analysis (Feb 27 – Mar 1)**
 - ✧ 2.1.1 Identify task characteristics
 - ✧ 2.1.2 Identify technology characteristics
 - ✧ 2.1.3 Calculate weighted TTF score
- **2.2 Conduct ECTM evaluation (Mar 2 – Mar 4)**
 - ✧ 2.2.1 Identify crucial use case
 - ✧ 2.2.2 Evaluate governance dimensions
 - ✧ 2.2.3 Calculate ECTM score
- **2.3 Assess governance and compliance risks (Mar 5 – Mar 7)**
 - ✧ 2.3.1 Review data privacy considerations
 - ✧ 2.3.2 Evaluate ethical impact
 - ✧ 2.3.3 Define mitigation strategies

3. System Design Phase (Mar 8, 2026 – Mar 26, 2026)

- **3.1 Define system architecture (Mar 8 – Mar 13)**
 - ✧ 3.1.1 Define LLM integration approach
 - ✧ 3.1.2 Define input processing module
 - ✧ 3.1.3 Define output formatting structure
- **3.2 Design user interaction workflow (Mar 14 – Mar 20)**
 - ✧ 3.2.1 Design voice input process
 - ✧ 3.2.2 Design text input alternative
 - ✧ 3.2.3 Define conversational flow logic
- **3.3 Design feedback generation logic (Mar 21 – Mar 26)**
 - ✧ 3.3.1 Define evaluation criteria
 - ✧ 3.3.2 Structure follow-up question generation
 - ✧ 3.3.3 Define performance feedback categories

4. Prototype Planning & Testing Framework (Mar 27, 2026 – Apr 16, 2026)

- **4.1 Define pilot testing framework (Mar 27 – Apr 2)**
 - ✧ 4.1.1 Define control and experimental groups
 - ✧ 4.1.2 Define testing timeline
 - ✧ 4.1.3 Establish evaluation rubric
- **4.2 Develop evaluation metrics (Apr 3 – Apr 9)**
 - ✧ 4.2.1 Define confidence measurement scale
 - ✧ 4.2.2 Define performance improvement metrics
 - ✧ 4.2.3 Define qualitative feedback collection

- **4.3 Design comparison study model (Apr 10 – Apr 16)**

- ✧ 4.3.1 Establish baseline performance
- ✧ 4.3.2 Define comparison methodology
- ✧ 4.3.3 Document expected outcomes

5. Final Review and Documentation (Apr 17, 2026 – Apr 30, 2026)

- **5.1 Compile final documentation (Apr 17 – Apr 22)**

- ✧ 5.1.1 Finalize all project reports
- ✧ 5.1.2 Review consistency across documents
- ✧ 5.1.3 Edit and format deliverables

- **5.2 Conduct final sponsor review (Apr 23 – Apr 26)**

- ✧ 5.2.1 Present final documentation
- ✧ 5.2.2 Address sponsor feedback
- ✧ 5.2.3 Secure sign-off

- **5.3 Submit final project deliverables (Apr 27 – Apr 30)**

- ✧ 5.3.1 Prepare final submission package
- ✧ 5.3.2 Submit to course platform
- ✧ 5.3.3 Confirm completion

PROMPT:

Use this project proposal and FRS to create a work breakdown structure for the project to three levels of definitions. The project starts January 29, 2026 and ends April 30, 2026. Use milestones corresponding to the project objectives.

MASY GC 4100 Assignment 1B Final Project Proposal
MEMORANDUM Requirement Specification Memo – AI-Powered Oral Exam Platform

Appendix E - Risk Management Plan

Project

AI-Powered Oral Exam Platform for Scalable and Valid Academic Assessment

This project aims to design and develop an AI-powered oral examination system that enables students to answer questions via voice and receive automated scoring and feedback.

Risks

Number	Risk	Probability Score (1,2 or 3)	Impact Score (1,2 or 3)
1	Underestimation of development time leading to incomplete features	3	3
2	Technical integration issues between frontend, backend, and AI APIs	3	3
3	AI model generates inaccurate or inconsistent evaluation results	2	3
4	Limited experience with full-stack development slows progress	3	2
5	Bugs in audio recording or playback functionality	2	2
6	Data handling and storage issues affecting system performance	2	2
7	Internet/API dependency causing delays or failures	2	3
8	Poor user interface affecting usability and user experience	2	2
9	Scope creep (adding too many features beyond initial plan)	3	2
10	Insufficient testing leading to low-quality deliverables	2	3

Risk Matrix

		RISK (exposure)		
Probability (of occurrence)		1.Slight	2. Moderate	3. High
	1. Very Unlikely			
	2. Possible		5,6,8	3,7,10
	3. Expected		4,9	1,2

Contingency Plan

Risk	Description	Probability (1-3)	Exposure (1-3)	Contingency Plan
1	Underestimation of development time	3	3	Prioritize core features (exam flow, scoring) and reduce non-essential features; create weekly milestones to monitor progress
2	Integration issues across system components	3	3	Test each module independently first; use APIs with clear documentation; allocate buffer time for debugging
3	AI evaluation inaccuracies	2	3	Implement fallback scoring rules and manually test outputs; refine prompts and validate results with sample answers
4	Limited full-stack experience	3	2	Focus on minimal viable product (MVP); reuse existing templates and libraries to accelerate development
7	API or internet dependency failures	2	3	Cache responses where possible; prepare backup demo (recorded results or screenshots)
9	Scope creep	3	2	Strictly follow initial project scope; document and postpone additional features
10	Insufficient testing	2	3	Conduct structured testing for each module; simulate real user scenarios before final demo

Appendix F – Technology Trial Plan

Business Task and Objectives

The business task addressed by this project is improving the effectiveness and scalability of oral academic assessments in higher education. Traditional oral exams require significant instructor time and are difficult to scale for large classes. As a result, students often receive limited opportunities to practice oral responses and receive feedback.

The objective of this project is to evaluate whether an AI-powered oral exam platform can improve students' preparation and performance in oral assessments by providing automated questioning, evaluation, and feedback.

Specifically, the project aims to determine whether the technology can:

- Provide structured oral practice opportunities for students
- Improve students' confidence in answering conceptual questions verbally
- Provide consistent and scalable feedback compared to traditional preparation methods

The Hypothesis

By introducing the AI-Powered Oral Exam Platform as a preparation tool, we hypothesize that students who use the system will demonstrate measurable improvements in oral response quality and confidence compared to students using traditional study methods.

Specifically, we expect that students using the platform will:

- Improve their response quality scores by at least 20% compared to the control group
- Increase their self-reported confidence in answering conceptual questions by at least 25%
- Reduce the average response preparation time by approximately 15% during practice sessions

These improvements will be measured through AI-generated response evaluations and pre- and post-trial student surveys over the course of the technology trial.

Baseline Metrics and Control Group

Baseline Metrics

The following baseline metrics will be measured before the trial begins:

- Student confidence in answering oral questions (survey-based)
- Accuracy and completeness of answers to conceptual questions
- Time taken to formulate responses
- Student satisfaction with current study methods

Control Group

The control group will consist of students who prepare for oral exams using traditional methods such as:

- reviewing lecture materials
- practicing with peers
- studying notes and textbooks

These students will not use the AI system during the trial period.

The Trial

The technology trial will involve introducing the AI-powered oral exam platform to a selected group of students and comparing their performance with the control group.

Trial Duration

The trial will run for approximately four weeks during the academic semester.

Trial Structure

Two groups of students will participate:

Control Group

- Students use traditional study methods.

Intervention Group

- Students use the AI-powered oral exam platform to practice answering conceptual questions.

The system will simulate oral exam questions and generate follow-up questions based on the student's response.

Students will receive feedback on:

- clarity of explanation
- conceptual understanding
- completeness of response

The performance and experience of both groups will then be compared.

The Technology

The technology introduced in this trial is an AI-powered conversational system designed to simulate oral exam interactions.

The platform will:

- present conceptual academic questions to students
- accept voice or text responses
- analyze responses using a large language model
- generate follow-up questions that probe deeper understanding
- provide structured feedback on the student's response quality

The goal is to replicate the experience of an instructor-led oral exam practice session while allowing scalable and automated interaction.

Data Collection

Data will be collected using both quantitative metrics and qualitative feedback.

System Data

The system will automatically collect data including:

- number of practice sessions completed
- response length and response time
- AI evaluation scores of responses
- improvement across multiple practice attempts

Survey Data

Participants will complete surveys measuring:

- confidence in answering oral questions
- perceived usefulness of the system
- satisfaction with the learning experience

Qualitative Feedback

Students will also provide open-ended feedback regarding:

- usability of the system
- clarity of feedback
- perceived usefulness for exam preparation

Analysis of Results

The results of the technology trial will be analyzed by comparing the control group and the intervention group across the defined metrics.

The analysis will include:

- comparison of student confidence levels before and after the trial
- evaluation of improvements in response quality
- comparison of engagement levels between groups
- review of qualitative feedback from participants

Statistical comparisons will help determine whether the AI system provides measurable improvements in student preparation.

Findings and Recommendations

The findings of the trial will summarize the impact of the AI-powered oral exam platform on student preparation and learning outcomes.

The final report will include:

- comparison of results between control and intervention groups
- evaluation of improvements in confidence and response quality
- identification of strengths and limitations of the technology

Based on the findings, recommendations will be made regarding:

- further development of the platform
- potential deployment in larger academic settings
- integration of AI-based oral exam practice tools into learning environments

Documentation of LLM use

Using a technology trial plan template, design a proof-of-concept technology trial for an AI-powered oral exam platform that helps students practice answering conceptual questions through simulated oral exams and AI-generated feedback.

Use these documents:

EXEMPLAR Design of the Technology Trial First pass Money Alignment Project.docx

MEMORANDUM Technology Trial Plan Template.docx

Design of A Trial of a Business Innovation Involving a Technology Change.docx.

Appendix G - Status Reports

Project Status Report

Your Name: Danxia Hong

Project Title: AI-Powered Oral Exam Platform for Scalable and Valid Academic Assessment

Date of report: 2/26/2026

1. Project Status and Explanation:

Project Status Area	Status (RYG)	Explanation
1. Overall Project Status		The project is currently progressing according to the planned timeline. Initial project documentation, including the proposal, FRS, TFT, and ECTM analyses, has been completed. No critical risks have been identified at this stage.
2. Project Schedule		The project schedule remains on track. All deliverables have been submitted within the required timeframe. Minor adjustments were made to refine documentation clarity, but no schedule delays have occurred.
3. Project Deliverables		All required deliverables to date have been completed and submitted. Documentation aligns with rubric requirements and incorporates feedback where applicable.
4. Resources & Collaboration		Collaboration with the project sponsor has been smooth. Required documents were reviewed and signed as requested. No resource constraints are currently impacting progress.
5. Changes		No major scope changes have occurred.
6. Communication		Communication with the sponsor and instructor has been timely and effective. Feedback has been incorporated into subsequent deliverables.

For status above, indicate **Red**, **Orange**, or **Green**:

- **Red**: Critical issues, serious risks to project, significant intervention must occur to achieve success, potential for stoppage of project activity. Project slipping by 5+ days, and resources uncommitted to meet deliverables
- **Orange**: Some major issues, moderate risk to project, must monitor closely, some internal or/and external dissatisfaction with progress. Project plan slipping by 2+ days.
- **Green**: No major issues, minimal risk to project, on target with expected outcomes, project on schedule, everyone satisfied with progress.

2. List All Completed Project Tasks:

- Finalized Project Proposal
- Completed Functional Requirements Specification (FRS)
- Conducted Task-Technology Fit (TTF) Analysis
- Completed ECTM Evaluation
- Secured Sponsor Agreement Documentation

3. List any concerns or issues that need the professor's involvement:

- No major concerns currently. Guidance will be requested if scope expansion or technical implementation questions arise.

4. Next series of tasks to complete:

- Refine system prototype logic
- Develop pilot testing framework
- Prepare evaluation metrics for user testing
- Continue sponsor alignment

5. Sponsor Signoff

Sponsor indicates agreement with the above status report:

By (signature): Andres Fortino
Project Sponsor

Printed Name: Andres Fortino

Project Status Report

Your Name: Danxia Hong

Project Title: AI-Powered Oral Exam Platform for Scalable and Valid Academic Assessment

Date of report: 03/26/2026

1. Project Status and Explanation:

Project Status Area	Status (RYG)	Explanation
1. Overall Project Status		Core system functionality has been successfully implemented, including exam workflow, audio playback, and answer submission. Project is on track toward completion.
2. Project Schedule		Some delays occurred due to API quota limitations and debugging issues (database schema mismatch), but progress remains steady with adjusted priorities.
3. Project Deliverables		Major deliverables such as user authentication, exam creation, and student exam participation have been completed. Remaining features are mainly enhancements.
4. Resources & Collaboration		OpenAI API quota issue has been resolved, enabling full testing and integration of the AI-based evaluation module. System capabilities have been significantly improved.
5. Changes		Minor adjustments made, including prioritizing core features over advanced analytics and refining system architecture.
6. Communication		Regular communication maintained with instructor; no major communication issues encountered.

For status above, indicate **Red**, **Orange**, or **Green**:

- **Red**: Critical issues, serious risks to project, significant intervention must occur to achieve success, potential for stoppage of project activity. Project slipping by 5+ days, and resources uncommitted to meet deliverables
- **Orange**: Some major issues, moderate risk to project, must monitor closely, some internal or/and external dissatisfaction with progress. Project plan slipping by 2+ days.
- **Green**: No major issues, minimal risk to project, on target with expected outcomes, project on schedule, everyone satisfied with progress.

2. List All Completed Project Tasks:

- Implemented user registration and login system (student & teacher roles)
- Developed exam creation functionality for instructors
- Built backend database models and API endpoints
- Enabled students to view and start available exams
- Implemented question display with progress tracking
- Integrated audio playback for questions using ElevenLabs API
- Developed answer submission system (text + voice recording)
- Implemented exam flow (next question logic & completion handling)
- Integrated OpenAI API for answer evaluation (basic functionality)
- Stored answers, scores, and feedback in database
- Implemented total score calculation logic

3. List any concerns or issues that need the professor's involvement:

- OpenAI API quota issue has been successfully resolved; no further action required
- No major issues requiring professor's involvement at this time

4. Next series of tasks to complete:

- Fix duplicate feedback issue in exam results
- Complete "View Details" page (exam results display UI)
- Improve AI evaluation stability and consistency
- Implement speech-to-text integration (optional enhancement)
- Conduct system testing and debugging
- Finalize UI/UX improvements
- Prepare final demo and presentation

5. Sponsor Signoff

Sponsor indicates agreement with the above status report:

By (signature):

Andres Fortino

Project Sponsor

Printed Name:

Andres Fortino

Please print in English

Final Project Status Report

Your Name: Danxia Hong

Project Title: AI-Powered Oral Exam Platform for Scalable and Valid Academic Assessment

Date of report: 04/16/2026

1. Project Status and Explanation:

Project Status Area	Status (RYG)	Explanation
1. Overall Project Status	●	The platform has been successfully developed and verified through comprehensive simulated testing environments.
2. Project Schedule	●	All milestones were met on or before the scheduled dates, aligning with the end-of-April completion goal.
3. Project Deliverables	●	All primary deliverables, including the MVP, GitHub repository, and final documentation, have been produced.
4. Resources & Collaboration	●	Collaboration with Dr. Fortino was consistent, and the project remained within the 300-350 hour resource estimate.
5. Changes	●	Validation methodology was refined from a live pilot to a structured simulation to ensure data control.
6. Communication	●	Weekly progress reports were maintained, and all four required stakeholder meetings were scheduled/conducted.

For status above, indicate **Red**, **Orange**, or **Green**:

- **Red**: Critical issues, serious risks to project, significant intervention must occur to achieve success, potential for stoppage of project activity. Project slipping by 5+ days, and resources uncommitted to meet deliverables
- **Orange**: Some major issues, moderate risk to project, must monitor closely, some internal or/and external dissatisfaction with progress. Project plan slipping by 2+ days.
- **Green**: No major issues, minimal risk to project, on target with expected outcomes, project on schedule, everyone satisfied with progress.

2. List All Completed Project Tasks:

- System Architecture & Design: Finalized the multi-agent workflow and instructor-facing configuration modules.
- MVP Development: Successfully built the voice-based conversational interface and the grading synthesis engine.
- Simulated Validation Study: Conducted simulated oral exams using 10+ representative student profiles to test grading consistency.
- Grading Accuracy Calibration: Verified that the "council-of-LLMs" framework delivers feedback aligned with predefined course objectives.

3. List any concerns or issues that need the professor's involvement:

- There are no outstanding technical or administrative issues. The project is ready for final handover.

4. Next series of tasks to complete:

- Final Presentation: Present the platform's simulated performance and results to Dr. Fortino on 04/28/2026.
- Repository Finalization: Complete the documentation in the public GitHub repository, including the README user guide.
- Final Report Submission: Submit the comprehensive project report by the end of the semester.

Please document the end of the project checklist of deliverables.

Please document how you accomplished these items

☒ Did the you arrange *at least* four meetings with the client during the project:

☒ Initial meeting to launch project

DATES: 01/22/2026

☒ Second meeting no more than two weeks after launch to review objectives

DATES: 02/26/2026

☒ Third meeting to review progress no more than two months after launch

DATES: 04/16/2026

☒ Final meeting to present results and hand in deliverables.

DATES: 04/28/2026

☒ Have you or will you provide a final report conforming to the template provided by the client?

☒ Will you or have you provide a repository of all final project files and a README user document in a public GitHub repository?

- ☒ Did you send weekly progress reports in written and email formats? These reports should include a summary of what has been accomplished in the past week, what tasks will be tackled in the upcoming week, and any issues that require resolution and input from the client.

5. Sponsor Signoff

Sponsor indicates agreement with the above status report:

By (signature): Andres Fortino
Project Sponsor

Printed Name: Andres Fortino
Please print in English

Appendix H - Annotated Bibliography

1. Deepshikha, D. A comprehensive review of AI-powered grading and tailored feedback in universities. *Discov Artif Intell* 5, 251 (2025). <https://doi.org/10.1007/s44163-025-00517-0>

Ranking: 10/10

Abstract:

AI-powered grading systems can automate assessment, improve efficiency, and provide personalized feedback while reducing bias and workload in higher education.

Summary:

This paper reviews recent advancements in AI-based grading systems, highlighting how machine learning and NLP can automate evaluation and generate personalized feedback. It also discusses challenges such as bias, transparency, and ethical concerns.

Researcher Comments:

This is highly relevant to my project because it directly addresses AI-based grading systems. It supports the feasibility of automating oral exam evaluation and provides insight into challenges I must consider.

2. Pecuchova, J., Benko, L. & Drlik, M. Automated Grading of Open-Ended Questions in Higher Education Using GenAI Models. *Int J Artif Intell Educ* 35, 3813–3846 (2025). <https://doi.org/10.1007/s40593-025-00517-2>

Ranking: 9/10

Abstract:

Recent GenAI models demonstrate strong performance in grading open-ended responses and providing meaningful feedback in educational settings.

ChatGPT Summary:

This study explores how generative AI models can assess open-ended responses and provide structured feedback. It shows improvements in grading accuracy and scalability.

Researcher Comments:

This directly relates to my system's use of LLMs for evaluating student answers, especially in open-ended oral responses.

3. Chamberlain, C., et al. (2025). Seneviratne, H., Manathunga, S. Artificial intelligence assisted automated short answer question scoring tool shows high correlation with human examiner markings. *BMC Med Educ* 25, 1146 (2025). <https://doi.org/10.1186/s12909-025-07718-2>

Ranking: 9/10

Abstract:

AI-based scoring systems show high correlation with human grading, supporting their reliability in educational assessment.

ChatGPT Summary:

The paper demonstrates that AI scoring systems can match human graders with high accuracy, particularly in structured responses.

Researcher Comments:

This validates the reliability of AI grading, which is critical for my project's credibility.

4. Yan, L., Sha, L., Zhao, L., Li, Y., Martinez-Maldonado, R., Chen, G., Li, X., Jin, Y., & Gašević, D. (2024). Practical and ethical challenges of large language models in education: A systematic scoping review. *British Journal of Educational Technology*, 55, 90–112.
<https://doi.org/10.1111/bjet.13370>

Ranking: 9/10

Abstract:

LLMs can automate grading and feedback but raise concerns about transparency, bias, and ethical implementation.

Summary:

This paper reviews how LLMs are used in education and highlights key ethical and practical challenges such as fairness and trust.

Researcher Comments:

This helps me understand risks in AI grading and ensures my system design considers fairness and ethics.

5. Gao, R., Guo, X., Li, X., Narayanan, A. B. L., Thomas, N., & Srinivasa, A. R. (2024). Towards scalable automated grading: Leveraging large language models for conceptual question evaluation in engineering.
arXiv. <https://doi.org/10.48550/arXiv.2411.03659>

Ranking: 10/10

Abstract:

The study explores the use of LLMs for grading conceptual questions in engineering, aiming to bridge the gap between human-like understanding and automated scalability. By leveraging sophisticated prompt engineering and rubrics, the authors demonstrate that models like GPT-4 can provide consistent, high-quality evaluation and feedback, rivaling human experts in efficiency and reliability.

Summary:

This paper explores the use of large language models (LLMs) for automated grading of conceptual engineering questions. It demonstrates that LLMs can effectively evaluate open-ended responses by understanding underlying concepts rather than relying solely on keywords, improving both scalability and grading consistency.

Researcher Comments:

This paper explores the use of large language models (LLMs) for automated grading of conceptual engineering questions. It demonstrates that LLMs can effectively evaluate open-ended responses by understanding underlying concepts rather than relying solely on keywords, improving both scalability and grading consistency.

6. Marcus Messer, Neil C. C. Brown, Michael Kölling, and Miaoqing Shi. 2024. Automated Grading and Feedback Tools for Programming Education: A Systematic Review. *ACM Trans. Comput. Educ.* 24, 1, Article 10 (March 2024), 43 pages.
<https://doi.org/10.1145/3636515>

Ranking: 8/10

Abstract:

This study explores the evolution of automated grading tools in programming education, focusing on how immediate feedback loops impact student mastery of complex logical concepts. The research demonstrates that by providing real-time diagnostic reports, these tools allow educators to scale personalized support to large cohorts in a way that traditional manual grading cannot sustain.

Summary:

The paper reviews various automated assessment systems, concluding that instant, precise feedback significantly improves learning curves and increases student engagement by reducing the turnaround time between submission and evaluation.

Comments:

This supports the "Real-time Interaction" requirement of my prototype. By using conversational voice AI, my system can simulate this instant feedback loop during an oral exam, helping students clarify their thoughts while the interaction is still active.

7. Najafi, F. T., & Pabba, V. R., & Subramanian, R., & Vidalis, S. M. (2025, March), AI-Assisted Grading – A Study on Efficiency and Fairness Paper presented at 2025 ASEE Southeast Conference, Mississippi State University, Mississippi. 10.18260/1-2—54139

Ranking: 8/10

Abstract:

This research compares AI-assisted grading systems with traditional human markers within engineering education contexts. The data indicates that AI systems demonstrate superior consistency and reliability when evaluating open-ended responses, effectively eliminating the grade volatility often caused by human fatigue or subconscious bias.

Summary:

The study highlights that AI-driven grading not only offers massive efficiency gains in speed but also adheres more strictly to pre-defined rubrics, ensuring a fairer assessment environment across diverse student populations.

Researcher Comments:

This provides a strong justification for the "Council-of-LLMs" framework. It allows me to argue that an AI-driven oral exam is not just faster, but potentially more objective than a single human examiner.

8. Tan, L. Y., Hu, S., Yeo, D. J., & Cheong, K. H. (2025). A Comprehensive Review on Automated Grading Systems in STEM Using AI Techniques. *Mathematics* (2227-7390), 13(17).

Ranking: 9/10

Abstract:

This systematic review analyzes the integration of advanced AI techniques, such as transformer models and deep learning, into STEM-based automated grading systems. The authors evaluate how these systems process diverse, non-structured student responses across various formats, emphasizing the transition from simple pattern matching to complex semantic understanding. The paper also identifies critical challenges in multimodal evaluation, particularly in maintaining high accuracy when students provide creative or unconventional explanations in technical subjects.

Summary:

The study examines the landscape of AI grading in STEM education, highlighting how sophisticated algorithms now allow for more nuanced assessments of student logic. It underscores that while technology has advanced, the primary hurdle remains the ability of AI to interpret the "reasoning path" behind a student's answer rather than just the final output.

Researcher Comments:

This is vital for understanding how the "Multi-agent Workflow" in my project can handle non-structured voice inputs. It emphasizes that my prototype must go beyond simple keyword detection to truly validate "genuine student understanding" during an oral exam.

9. Liu, X., Xiao, Y. & Li, D. Assessing strategic use of artificial intelligence in self-regulated learning: Instrument development and evidence from Chinese university students. *Int J Educ Technol High Educ* 22, 69 (2025). <https://doi.org/10.1186/s41239-025-00567-5>

Ranking: 10/10

Abstract:

This study addresses the gap in measuring how university students strategically integrate AI into their self-regulated learning (SRL) processes. The authors developed and validated a multi-dimensional instrument that captures key behaviors, including AI-assisted goal setting, monitoring, and performance evaluation. By analyzing empirical evidence from Chinese university students, the research highlights that "strategic use" of AI—rather than passive consumption—is a critical predictor of academic success and cognitive engagement in AI-enhanced environments.

Summary:

The paper provides a comprehensive framework for assessing student agency in the age of AI, demonstrating that successful learners use AI as a scaffold for their own metacognitive processes. It introduces a validated scale that categorizes AI usage into strategic planning and reflection, emphasizing that the human-AI interaction must be intentional to yield genuine learning gains.

Researcher Comments:

This is a crucial addition to my project's theoretical framework, specifically for the Validation Study and UX Analysis. Since my platform requires students to engage in a high-stakes voice-based oral exam, this paper helps me understand how their ability to "self-regulate" with AI might influence their performance. Furthermore, it provides a benchmark for evaluating whether my system successfully encourages "strategic" responses from students rather than rote memorization or AI-generated scripts.

10. Kasneci, E., et al. (2023). Kasneci, E., Seßler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., ... & Kasneci, G. (2023). ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and individual differences*, 103, 102274.

Ranking: 8/10

Abstract:

This comprehensive position paper, authored by a multidisciplinary team of experts, provides a systematic overview of how Large Language Models (LLMs) can be harnessed to enhance educational environments while mitigating systemic risks. The authors explore the technical architecture of models like ChatGPT and their capacity for providing personalized tutoring, automated feedback, and sophisticated assessment support. Crucially, the paper outlines a roadmap for "educational transformation," emphasizing the need for new digital literacy skills and the redesign of traditional evaluation frameworks to accommodate AI's capabilities.

Summary:

The study serves as a foundational guide for integrating LLMs into higher education, arguing that AI can significantly reduce administrative burdens on faculty while offering students highly tailored learning experiences. It balances the "opportunities" (such as scalable 1-on-1 tutoring) against the "challenges" (such as academic integrity and algorithmic bias), concluding that institutional success depends on a proactive, policy-driven approach to AI adoption.

Researcher Comments:

This is the most "all-encompassing" citation for my project. It provides the high-level academic authority needed to justify my multi-agent grading framework. I use this paper to support the idea that AI-driven assessment is not just a defensive measure against cheating, but a "positive opportunity" to provide NYU SPS students with more frequent, detailed, and personalized feedback than a human instructor could manually provide.

11. Cotton, D. R., Cotton, P. A., & Shipway, J. R. (2024). Chatting and cheating: Ensuring academic integrity in the era of ChatGPT. *Innovations in education and teaching international*, 61(2), 228-239.

Ranking: 8/10

Abstract:

This research investigates the profound disruption caused by Large Language Models (LLMs) in higher education, specifically focusing on the vulnerability of traditional written assessments. Through a series of experiments, the authors demonstrate that ChatGPT can produce high-quality academic content that easily bypasses current plagiarism detection software, creating a crisis for institutional academic integrity. The paper concludes that universities must pivot away from static, take-home written tasks and instead adopt "authentic assessment" models that prioritize real-time student interaction, critical thinking, and the oral defense of ideas.

Summary:

The study highlights an "academic arms race" where technological advancements are

outpacing institutional regulations. It warns that relying on AI-detection tools is a failing strategy and advocates for a fundamental redesign of the assessment landscape. The authors suggest that evaluations must become more process-oriented and interactive to ensure that a student's grade reflects their true cognitive effort rather than their ability to prompt an AI.

Researcher Comments:

This is the cornerstone of my "Problem Statement" for the NYU SPS project. It provides the definitive academic argument that written exams are no longer a reliable measure of student knowledge. I cite this to justify the technical necessity of my AI-Powered Oral Exam Platform, framing it as the exact type of "authentic, interactive assessment" that the authors recommend to safeguard academic integrity in the 21st century.

12. Binns, R. (2018). Algorithmic accountability and public reason. *Philosophy & technology*, 31(4), 543-556.

Ranking: 8/10

Abstract:

This paper explores the intersection of machine learning and political philosophy, specifically the concept of "public reason" as a requirement for algorithmic accountability. Binns argues that when automated systems make high-stakes decisions, such as grading or professional evaluation, they must be able to provide justifications that are transparent, evidence-based, and understandable to the individuals affected. The research suggests that accountability is not just about technical accuracy, but about the system's ability to offer a "reasoned defense" of its outcomes in a way that respects the autonomy of the user.

Summary:

The study moves beyond the "black box" problem of AI, proposing that algorithms should be designed to mirror the standards of human public reasoning. It emphasizes that for an AI system to be considered legitimate in a professional or academic setting, its decision-making process must be articulable and open to scrutiny, ensuring that the logic behind a specific result can be traced and debated.

Researcher Comments:

This paper provides the theoretical foundation for the "Evidence-based Feedback" feature of my platform. Since my prototype uses a "Council-of-LLMs" to grade oral exams, Binns' work justifies why the system must output a detailed "deliberation transcript" alongside the final grade. It ensures that the project meets high standards of algorithmic accountability, providing NYU SPS students with a clear, reasoned explanation for their assessment outcomes rather than an arbitrary score.

13. Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. 2021. A Survey on Bias and Fairness in Machine Learning. *ACM Comput. Surv.* 54, 6, Article 115 (July 2022), 35 pages. <https://doi.org/10.1145/3457607>

Ranking: 8/10

Abstract:

This extensive survey provides a taxonomy of the various types of bias that can infect machine learning systems, tracing their origins from data collection and representation to model training and deployment. The authors identify over 20 distinct types of bias including historical, representation, and measurement bias, and review state-of-the-art fairness definitions and mitigation strategies. The paper emphasizes that as AI systems are increasingly used for societal decision-making, understanding the mathematical and social nuances of "fairness" is essential to preventing the reinforcement of existing inequalities.

Summary:

The research offers a holistic view of the "bias problem" in AI, categorizing how errors in data and algorithmic logic can lead to systematically unfair outcomes for specific groups. It provides a roadmap for researchers to implement fairness-aware interventions, suggesting that both pre-processing data and post-processing model outputs are necessary to ensure equitable performance across diverse user demographics.

Researcher Comments:

This survey is a critical resource for the Validation and Evaluation phase of my project. Because my platform conducts oral exams for a global student body at NYU SPS, I must ensure that the "Council-of-LLMs" does not exhibit linguistic or cultural bias against students with different accents or dialectical variations. I use the frameworks provided in this paper to design my bias-mitigation testing, ensuring that the automated grading system remains equitable and fair for all students regardless of their background.

14. Holmes, W., Porayska-Pomsta, K., Holstein, K. et al. Ethics of AI in Education: Towards a Community-Wide Framework. *Int J Artif Intell Educ* 32, 504–526 (2022).
<https://doi.org/10.1007/s40593-021-00239-1>

Ranking: 9/10

Abstract:

This influential paper addresses the urgent need for a comprehensive ethical framework governing the use of Artificial Intelligence in Educational (AIEd) settings. The authors argue that AI ethics in education must move beyond narrow technical considerations of data privacy and algorithmic bias to include broader pedagogical and social concerns, such as the impact of AI on the teacher-student relationship and the potential for "automated" inequality. The study proposes a community-wide framework that involves collaboration between developers, educators, and students to ensure that AI tools are designed and deployed in ways that are transparent, accountable, and aligned with human-centric educational values.

Summary:

The research highlights that as AI becomes more integrated into classrooms, "ethics" must be treated as a continuous process rather than a one-time checklist. It advocates for an "Ethics-by-Design" approach, where pedagogical integrity and student well-being are prioritized during the development phase. The paper emphasizes the importance of stakeholder participation, ensuring that the voices of those most

affected by AI instructors and learners are central to the development of ethical guidelines and institutional policies.

Researcher Comments:

This paper is a direct influence on my Stakeholder Alignment and Implementation Strategy. It justifies my approach of working closely with Dr. Andres Fortino (the project sponsor) to ensure the AI oral exam platform respects the "pedagogical relationship" at NYU SPS. I use this framework to argue that my project is not just a technical tool for assessment, but an ethically grounded system designed to enhance not replace the instructor's role by providing them with deeper, more transparent insights into student learning through a "community-validated" approach to AI ethics.

15. Zawacki-Richter, O., Marín, V. I., Bond, M., & Gouverneur, F. (2019). Systematic review of research on artificial intelligence applications in higher education—where are the educators?. *International journal of educational technology in higher education*, 16(1), 39.

Ranking: 9/10

Abstract:

This highly influential systematic review maps the landscape of AI research in higher education, identifying four main areas of application: profiling and prediction, assessment and evaluation, adaptive systems and personalization, and intelligent tutoring systems. The authors reveal a significant imbalance in the literature, noting that while technical possibilities are widely explored, there is a lack of focus on pedagogical reflection and the actual involvement of educators in the design of these systems. The paper concludes with a call for more "human-centric" AI applications that prioritize the teacher's expertise and the student's learning experience over mere technical automation.

Summary:

The study provides a comprehensive taxonomy of AI applications in universities, highlighting that most current research is driven by computer science rather than educational science. It critiques the "black box" nature of many AI tools and emphasizes that for AI to be truly effective in higher education, it must be designed with pedagogical integrity and involve faculty members as active stakeholders in the assessment process.

Researcher Comments:

This paper is the primary source I use to justify the "Instructor Configuration" and "Analytics Dashboard" components of my platform. By addressing the question "where are the educators?", my project ensures that the AI-powered oral exam is not an autonomous black box, but a tool controlled and tuned by NYU SPS faculty to meet specific course objectives. This citation helps me frame my project as a direct response to the gaps identified by Zawacki-Richter et al., ensuring that my "Council-of-LLMs" framework remains educationally grounded and instructor-led.